



Business/Data Roles in Technology: Skills for Recruiting and Career Growth

Delivered by Ray Wang & Bianca Bhardwaj, Breakout Classroom 1130

Meet your Workshop Leaders

Glad to be here!



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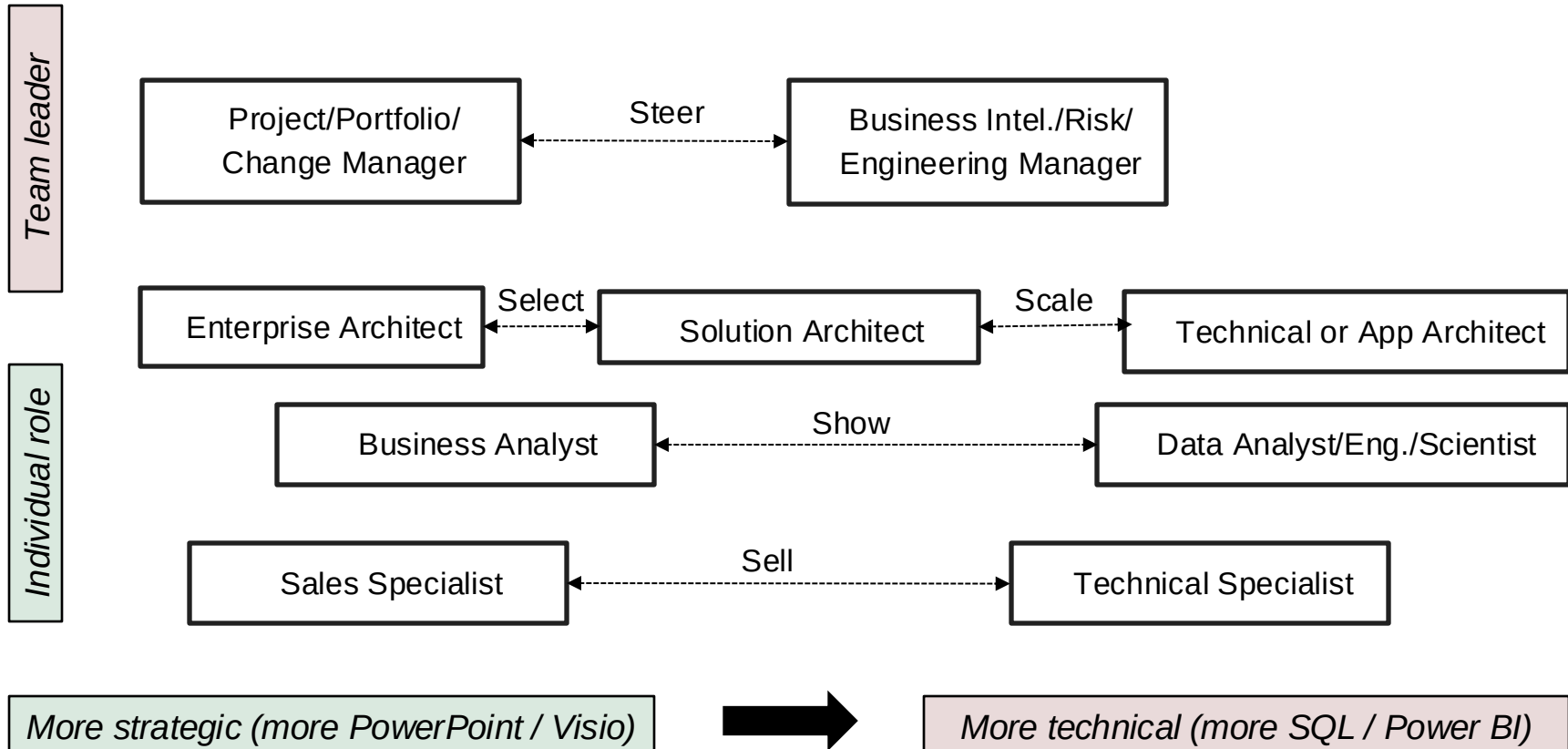


Section I: Introducing the Roles

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I want to work in tech, but I have a business background...

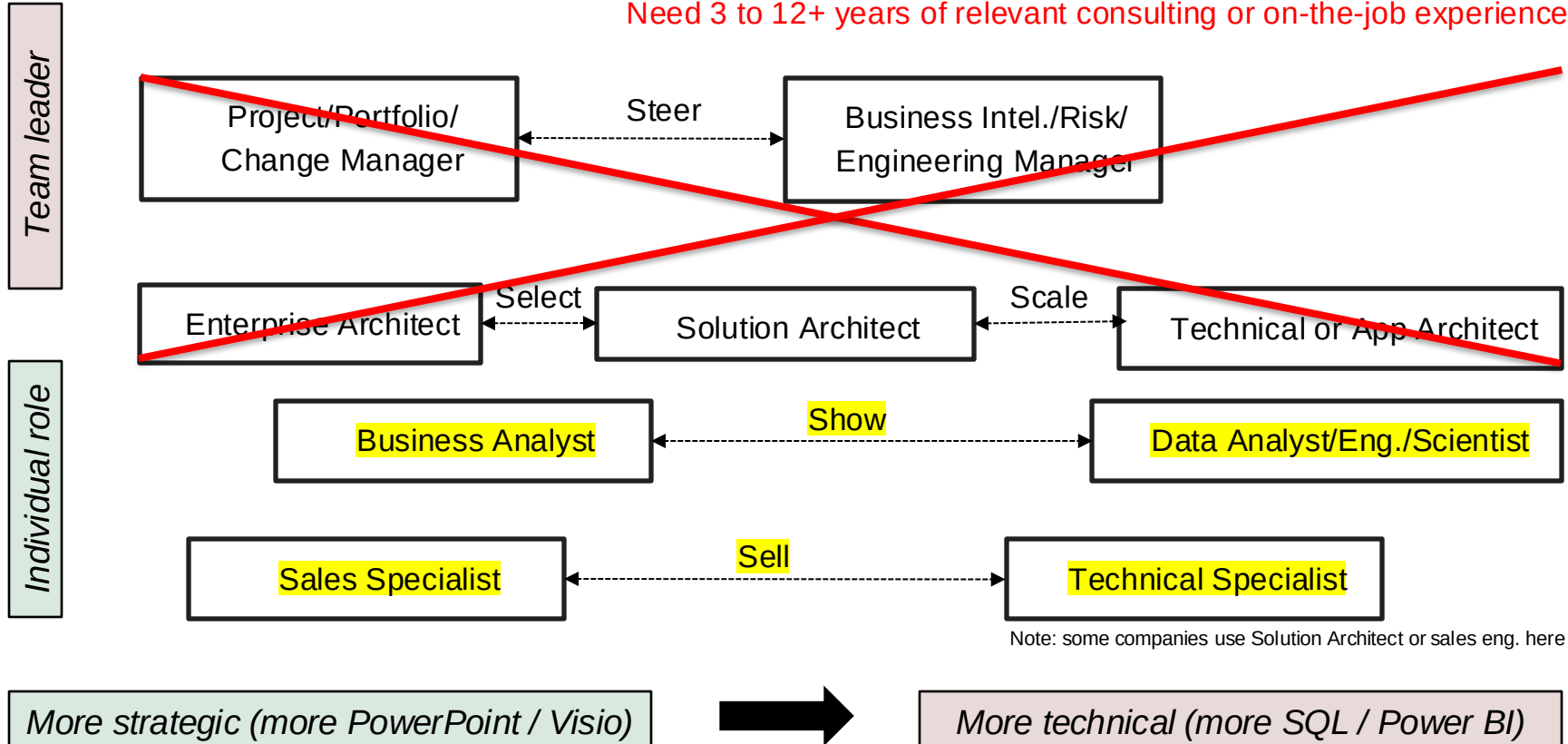
What business-heavy roles exist on technology teams?



Okay, but what can I do as a new grad?

Which roles are open to you on 12Twenty, LinkedIn, and company job boards?

Need 3 to 12+ years of relevant consulting or on-the-job experience



Role Brief: Sales Team Members

What do sales specialists and technical specialists do?

Sample Job Description

Sales team members design, pitch, and orchestrate the delivery of complex technology solutions for clients. Operating for a service provider such as AWS, Azure, GCP, or third parties like system integrators, they draw from their expert knowledge of a technology stack and their understanding of clients in a territory to guarantee the business and technical feasibility of product consumption. These team members may be split into **pre-sales** teams who teach and convince customers to make a sale happen, and **post-sales** teams focusing on satisfaction and repeated use.



TL;DR:

Sales specialists find and win sales for a tech company by making a strong business case. **Technical specialists** are experts on their products and help fit them into systems/problems.

Role Brief: Business Analyst

What does a Business Analyst do?

Sample Job Description

Business analysts (BAs) are crucial in bridging the gap between a company's business objectives and operational capabilities, enabling them to increase efficiency, reduce costs, and excel in performance. Frequently found working alongside technical team members, BAs draw from their deep understanding of systems to prioritize technical and functional requirements and drive forward feasible solutions that satisfy client needs. BAs are expected to confidently speak to developers and business stakeholders alike.



TL;DR:

Business analysts (BAs) fix processes and write logic, and acts as a liaison between various parties.

Role Brief: Data Analyst

What does a Data Analyst do?

Sample Job Description

Data analysts play an important role in helping organizations to make data-driven decisions that can improve operational efficiency, customer experience, and business outcomes. They use statistical methods, software tools, and programming languages to clean and manipulate data, create visualizations and dashboards, and develop predictive models. Data analysts must have strong analytical and problem-solving skills, as well as expertise in data visualization, data mining, and statistical analysis.

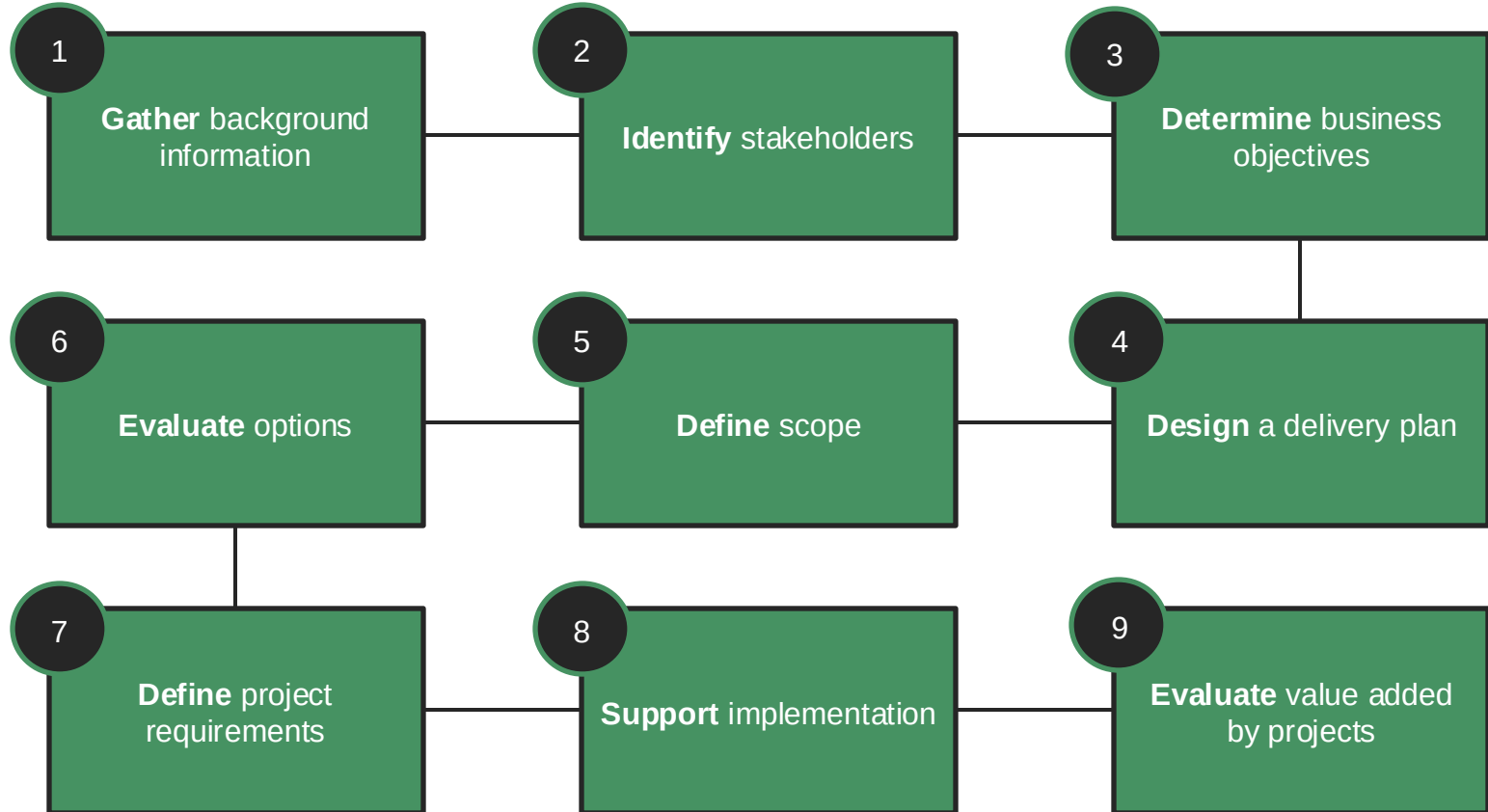


TL;DR:

Data analysts turn data into information by structuring, visualizing, and modelling with it.

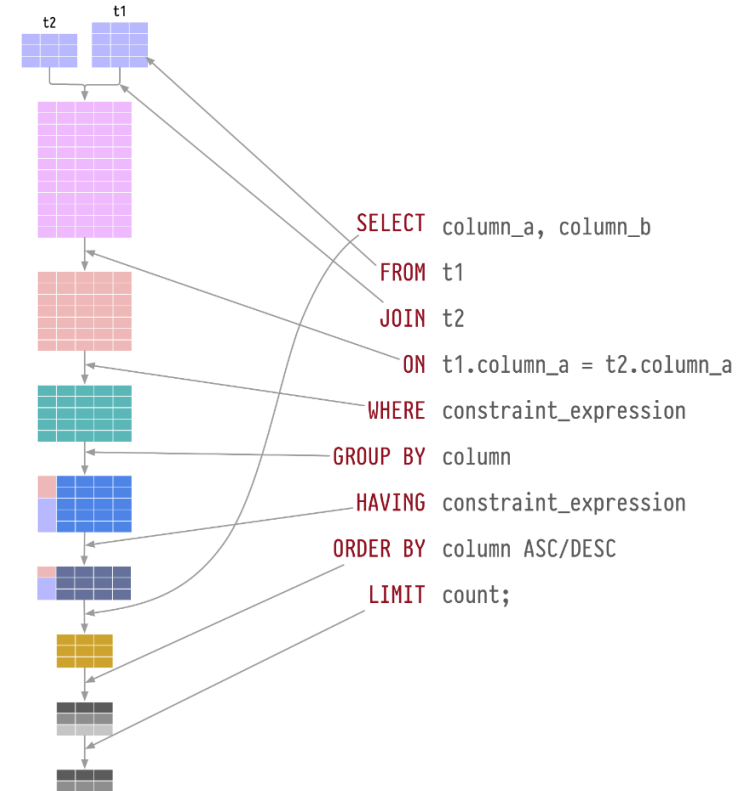
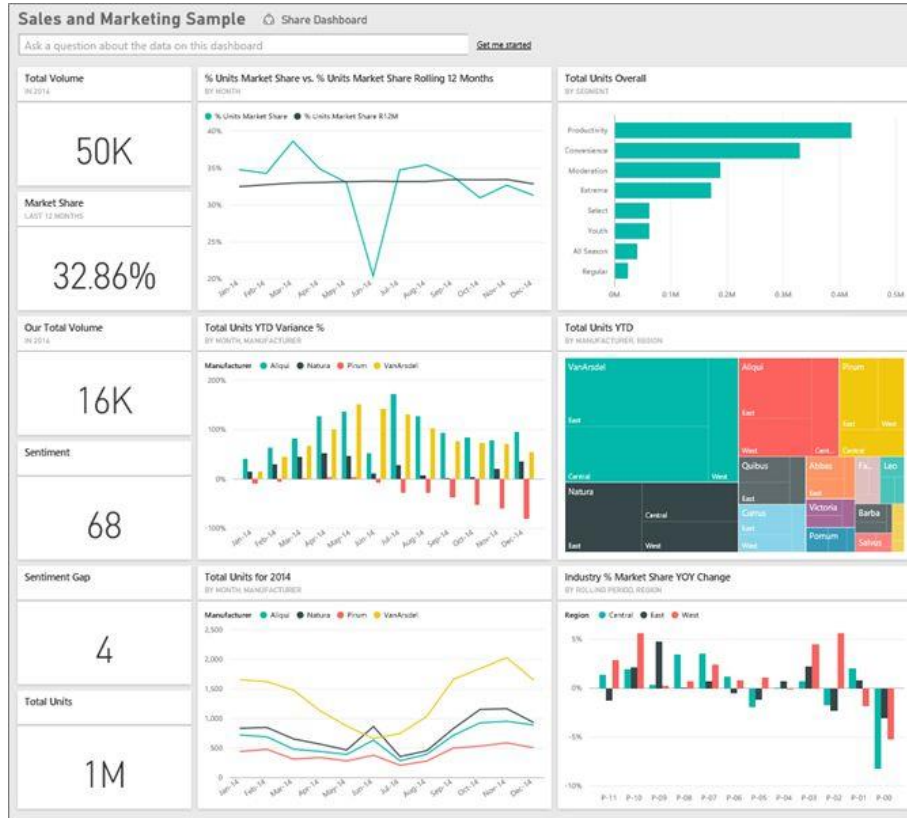
Work Samples: Business Analyst

A General Workflow



Work Samples: Data Analyst

Dashboarding and Filtering Logic

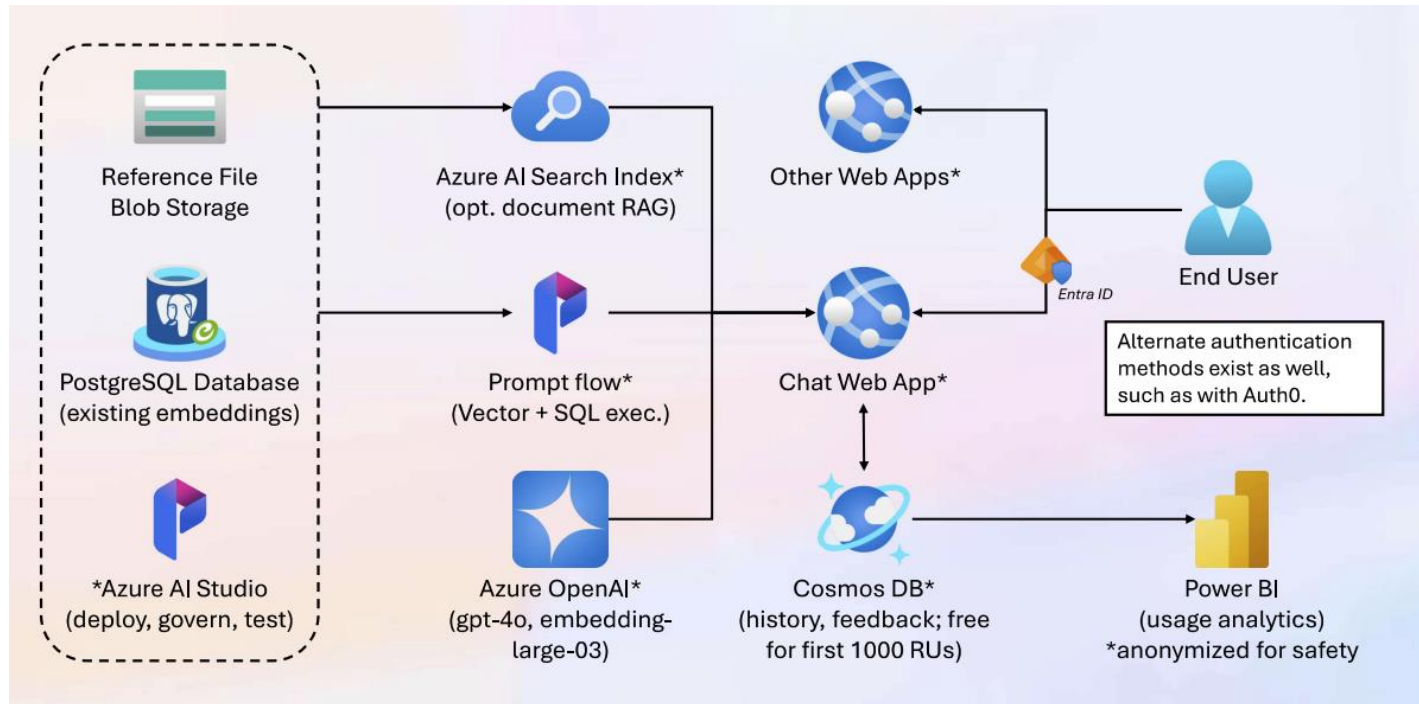


Work Samples: Sales / Technical Specialist

Application Architectures and Cost Projections

Sales Specialist: “I found that [public sector agency] wants a chatbot to search through their data.”

Technical Specialist: “Try this set up. I’ll get a working demo to show their engineering team next week.”





Section II: Key Skills and Competencies

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Top Skills: Business Analyst

Development and Presentation

You're **familiar** with ...

- SQL
- Statistical languages (R, Python)
- Visualization Tools (Power BI, Tableau)
- Analytics
 - Descriptive (What?)
 - Diagnostic (Why?)
 - Prescriptive (Action)
 - Predictive (Future)

You **are** ...

- A clear communicator
- Curious
- A problem-solver
- Detail-oriented
- A big picture thinker
- A negotiator

You can **build** by ...

- Taking courses
- Attending workshops
- Shadowing professionals
- Gaining work experience
- Finding a mentor

Top Skills: Data Analyst and Sales Team Roles

Development and Presentation

Key Role Skills

How to Develop

Showcasing your Strengths



1

Technical Fluency
(SQL, Python, Visuals)



Take Free Online Courses
([Power BI](#))



Get Certified (see next slide)
([Power BI](#))

Certifications and Bootcamps: Business/Data Roles in Technology



Cutting through the noise: are any worth it?

Certification/Course	Preparation	Worth It? + Benefit
1 Microsoft Power BI (Visuals and Wrangling)	Free Course on MS Learn (Skim Content, Cram Past MCQs)	YES (DATA) : 82.5USD or ~112CAD (FREE badges per module, 2 weeks to prep, good hands-on lab experience)
2 PMI CAPM (Intro to Project Mgmt.)	Paid Course + Exam PocketPrep (Power BI)	YES (BA) : 30 + 30 + 350CAD (25 hours of classes, 2 months to exam prep, very well-respected by project mgrs.)
3 Azure/GCP/AWS (Intro Cloud/Data/AI/CRM)	Free Course on Learn/equiv. (AZ-900, DP-900, AI-900, MB-910)	YES (ALL) : 56USD or ~76CAD (FREE badges per module, 2 weeks to prep, good hands-on lab experience)
4 Database Querying (sqlbolt)	Free online course program	YES (DATA/BA) : Totally free website with a very easy to use sandbox to grasp the fundamentals of SQL

Top Skills: Data Analyst and Sales Team Roles

Development and Presentation

Key Role Skills

1

Technical Fluency
(SQL, Python, Visuals)

2

Storytelling with Data

How to Develop

Take Free Online Courses
([Power BI](#))

Work with Sample Datasets
([Power BI](#))

Showcasing your Strengths

Get Certified (see next slide)
([Power BI](#))

Create an End-to-End Project
([Power BI](#))



Top Skills: Data Analyst and Sales Team Roles

Development and Presentation

Key Role Skills

1

Technical Fluency
(SQL, Python, Visuals)

2

Storytelling with Data

3

Pitching/Presenting

How to Develop

Take Free Online Courses
([Power BI](#))

Work with Sample Datasets
([Power BI](#))

Showcase/Interpret Projects

Showcasing your Strengths

Get Certified (see next slide)
([Power BI](#))

Create an End-to-End Project
([Power BI](#))

Participate in a Competition
(Articles/YouTube videos work too!)



Creating a Data Project

Development and Presentation



- **Historically**, it was difficult to gain hands-on experiences with data technologies if you didn't have access to a professionally-configured workspace through your job.
- **Today**, we experience a plethora of services that can be used extensively in a **free-tier**.

Some favourites of mine:

- ✓ Supabase (<https://supabase.com/>): totally free online Postgresql database, no install required
The best way to practice loading data, creating tables, and writing SQL!
- ✓ Microsoft Azure (<https://portal.azure.com>): free credits for students, can run Python scripts for free
If you want to build and test something in the cloud, you can do so with minimal cost!

How to stand out as a data candidate:

1. Pick a problem that you're very passionate about involving data collection and presentation; for example, grocery store prices in Canada.
2. Find a way to manually collect and store this data (spreadsheet/database); e.g., right from metro.ca
3. Automate the collection (*ingestion*) process using a Python script; e.g., web scraper (use ChatGPT!)
4. Write SQL to draw initial insights from your raw data and transform it to be more usable.
5. Neatly present your findings. Make visuals in Excel/Power BI, create a slide deck with findings, submit to a conference or participate in a case comp... or even launch a basic app that uses your data!



Section III: Early Career Progression

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Sample Early Career Path: Business Analyst

Responsibilities, timeline, and timeframes



BUSINESS ANALYST II – \$100,000

2-5 years

First promotion

Independently manage complex analysis projects, providing strategic recommendations to technical leads and senior management. Play a pivotal role in leading operational/strategic initiatives in the business unit.

BUSINESS ANALYST I – \$85,000

2-3 years

Post graduation

Support senior analysts and workstream leads in conducting process analyses, developing key artefacts (e.g. project management plan, feature backlog), and implementing process improvement initiatives.

BUSINESS ANALYST INTERN – \$70,000

4 months

HBA 1/HBA 2 summer

Work alongside a BA team, or a team of interns, to define user requirements, analyze implications for implementation, and document decisions and changes as required.

Sample Early Career Path: Data Analyst

Responsibilities and timeframes



SENIOR DATA ANALYST – \$130,000

2-5 years

First promotion

Lead data-driven projects, developing and training complex models for custom business applications. Provide recommendations on data strategy for new infrastructure/applications and mentor juniors.

DATA ANALYST – \$100,000

2-3 years

Post graduation

Develop applications to assist with data collection, processing, and analysis, such as predictive models and automated pipelines. Assist with BI tasks like reporting and data-driven strategy generation.

DATA ANALYST INTERN – \$70,000

4 months

HBA 1/HBA 2 summer

Support the data team with analysis, from preprocessing to modelling. Automate manual processes and develop ad-hoc solutions such as web scrapers to accelerate business/development velocity.

Sample Early Career Path: Sales/Technical Specialist

Responsibilities and timeframes



SENIOR SPECIALIST – \$180,000

2-5 years

First promotion

Proactively identify and win sales opportunities. Own and lead the development of complex architectures, aligning stakeholder and client objectives with selected technologies. Lead junior team members.

SPECIALIST – \$125,000

2-3 years

Post graduation

Learn and apply principles in both foundational architecture and your firm's specialized technology stack to design and implement new solutions. Lead sales calls, users, and managers to refine requirements.

SPECIALIST INTERN – \$60,000

4 months

HBA 1/HBA 2 summer

Gain familiarity with client need analysis and technology solution design by pursuing field-specific certifications while building proof-of-concepts for demos and internal users. Support client teams with tasks as assigned. Depending on sales/technical team assignment, focus on prospecting/building.



Section IV: Recruiting Timelines

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Recruiting Timelines

When should you apply?



August-September
Networking / Crafting
Resume



September-December
Mocking / Preparing
for Interviews



October/March
14-28 Days:
Second and Third
Round Interviews



September-December
Applications Open,
Apply + Get Referrals

October/March
7-14 Days:
Recruiter Phone
Screen / Take
Home
Assessments

21-28 Days
Offer!



Section V: Résumé, Interview, and Assessment Prep

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Resumé Breakdown

What goes into a good resumé?



Resume Breakdown

Name & Contact Information

Education

Professional Experience

Leadership Experience & Projects

Technical Skills & Interests

- Make your experiences **accomplishment-oriented**
 - 3-4 bullet points for each work experience or extracurricular experience
 - Quantify them as much as possible
 - Smartly structure your responses (next slide)
- Analyst-related Roles: Highlight **communication skills, taking initiative, problem-solving** and your ability to **empathize with customers**
 - *Product knowledge*
 - *Client acquisition and retention*
 - *Goal setting and forecasting*
 - *Self-motivation*
 - *Organization*
 - *Time management*
 - *Attention to detail*

Interview Process

Initial Screen and First through Final Rounds



1

Initial Phone/Video Screen (Recruiter or Associate)

- Key question of the interview: **Does this person meet the job requirements?**
- Vibe check, is the person knowledgeable but also down to earth?
- First interviews are all about strong communication and how aligned you are with the job description, plus any logistical questions (need visa sponsorship?)
- **Know the company and product in and out, the more specific the better**
- **Tell structured, concise, and relevant stories that relate to the work**

2

First, Second, and Third Round Interviews

- Behavioral questions (experience, leadership, failures)
- Maybe: Take home assignment/case
- Maybe: Project showcase (completed past work)
- Maybe: Technical questions (general data questions and SQL)

Bullets and Behavioural: Bread and Butter

Why STAR is lame, and how to do better on your resumé and interview



The O-A-I Resume Bullet Point Style

Output: your direct winning impact on the company itself; the value you captured.

Action: your skills and familiar technologies proven by this experience.

Input: your key achievement/measured success that enabled the winning impact.

Note: You most definitely don't have to follow this order. Having 3 elements where 2 focus (OI in OAI) on successes rather than 1 in 4 (R in STAR) is better.

Worked Example

Sales (OAI—Good Example):

- Cemented Databricks adoption and use at a major transit agency by developing a T-SQL to Spark SQL code converter tool, automating code conversion to save data migration team 80 hours over the course of two months.

Data (STAR—Bad Example):

- Developed chatbot for the federal government, saving business analysts time and presenting design to leadership for further use.

Lame, wordy without saying much, and non-quantified (overly qualitative).

Bullets and Behavioural: Bread and Butter

Why STAR is lame, and how to do better on your resumé and interview

The TSTARRS Behavioural Answer Style

Teaser/Result: your direct winning impact on the company itself.

Sit./Task/Action: your skills and familiar technologies proven by this experience.

Reflection/Strengthen: your ability to grow from any experience.

Note: Try to conclude by referencing your big win once again, but with great humility and deference to all of the exploration you had in the process.

What in the world is a TSTARRS?

- **Teaser:** Hook their attention, citing a massive result.
- **Situation:** Outline the problem statement and set the context.
- **Task:** Describe your role and circumstance, plus any overarching goals or people involved? Be concise—no one likes long set-ups.
- **Action:** Walk through step-by-step, including collaborations, difficult decisions, and seeking data or help. Build tension.
- **Result:** Demonstrate, proudly and measurably, the impact on the team and greater organization? Quantify! Teaser teases this.
- **Reflection:** Look back and find learnings or opportunities to improve—what did the experience tell you about the world?
- **Strengthen:** Preach how you can use your strengths and newfound learnings to address future situations.

Bullets and Behavioural: Bread and Butter

Why STAR is lame, and how to do better on your resumé and interview



Worked Example: TSTARRS / Final Round Microsoft Sales Interview

Western: WriteNow and the Microsoft for AI Accessibility Hackathon

Teaser:

- Here's the story of how I built a product that helped 650 Western accessible-needs students learn better and ace their finals...

Situation and Task:

- As someone whose close friends qualify for learning accommodations, I knew the system was broken; Western relied on an unreliable network of volunteers for lecture transcripts—1,000 emails go out per year. I wanted to automate this.

Action:

- New OpenAI tech: Whisper model; local prototype => testers => feedback rapidly from Oct. '22 launch → API Mar. '23
- Interviewed 100+ students, found opportunities to pitch to CGI U—deans, changemakers, developers
- Expanded team to include devs, pivoted to the cloud => refactored codebase weekly to use new APIs for Whisper and ChatGPT as they became available, pivoting focus

Bullets and Behavioural: Bread and Butter



Why STAR is lame, and how to do better on your resumé and interview

Worked Example: TSTARRS / Final Round Microsoft Sales Interview

Western: WriteNow and the Microsoft for AI Accessibility Hackathon

Result:

- Built a robust product, pitched and won at the Microsoft AI for Accessibility challenge, using our \$8,000 in prize money to pay for our API calls and improve the app for finals
- Over 650 accessible-needs users signed up, 1 in 4 of those with accommodations at Western, average student was able to score >90% on their finals

Reflection:

- ChatGPT expanded to include our functionality, Western now offers generated transcripts by default, we ended our work—it's hard to beat innovation pace of a giant (built-in transcript / RAG patterns beat our hacky-solution)
- HOWEVER, I learned that if I iterate and build quickly, I can deliver effective proof-of-concepts and sell a population on a solution or design pattern before it's truly "fully built out" → THE WORK OF A PRE-SALES TECHNICAL SPECIALIST

Action:

- This was my launching point into working with generative AI and LLMs (we were quite literally the first to get access to the APIs to build with the product) safely and productively—did this successfully at EY, and now hopefully Microsoft!



Section VI: How to Stand Out in the Next Month and Year

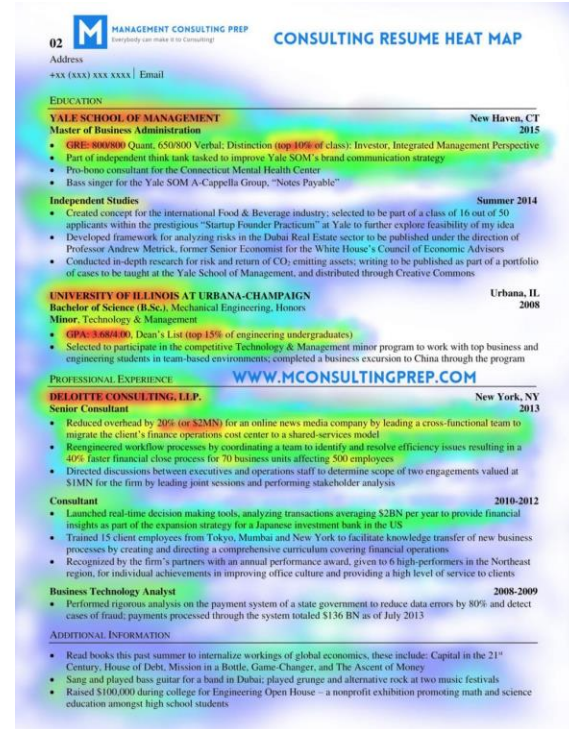
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Differentiating Yourself as a Candidate

The clock is ticking...

Within one month, you should...

1. Rewrite your entire resume in **OAI format**, cutting all irrelevant points.
2. Brainstorm and rewrite your core behavioural stories in **STARRS** format.
3. Brush up on any core skills which you are missing using the free resources.
4. Complete at least one certification related to your desired BizOps field.
5. Clean up a past project (school or otherwise) to showcase.



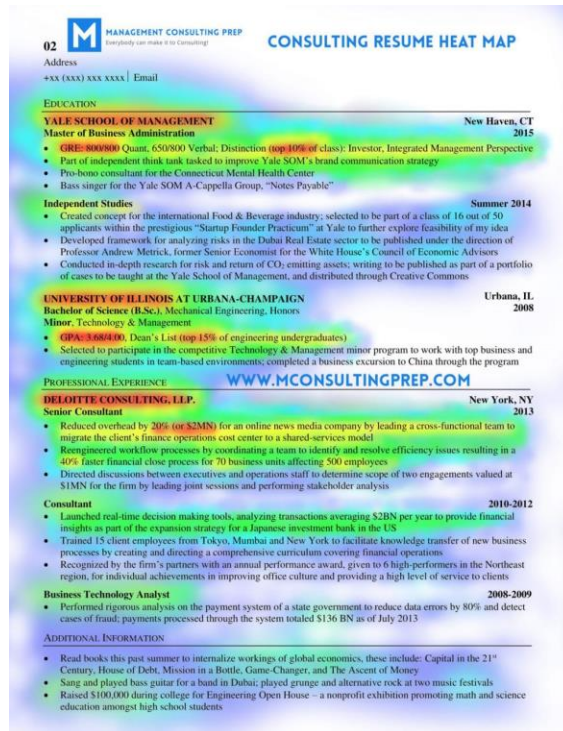
The resume heatmap clearly shows that the OAI method is OP. Recruiters won't even see results in STAR format (not reading all that).

Differentiating Yourself as a Candidate

The clock is ticking...

Within one year, you should...

1. **Seek internship and part-time work** that best fits your field of interest.
2. **Design and deliver** your very own data project, hopefully attaining some users.
3. **Pitch and showcase** your work at a conference or competition.
4. Complete at more or deeper certification related to your desired BizOps field.
5. Enjoy SQL Leetcode. Speaking of which...



The resume heatmap clearly shows that the OAI method is OP. Recruiters won't even see results in STAR format (not reading all that).

SURPRISE LEETCODE

ALWAYS BE ON GUARD (thanks SWE friends)



Meet your Workshop Leaders

It was a blast, let's stay in touch!



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Bianca Bhardwaj
**Prev. Finance Modernization
at CIBC**

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I taught English in Costa Rica!
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Thank you!

Questions, next steps?



Product Management Fundamentals: Skills for Recruiting and Career Growth

Delivered by Bruce Liu, Rachel Zhu, Grace Zhou, Sophia Ma, Breakout Classroom 1200

Meet your Workshop Leaders

We're super excited to be here!



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Section I: Introducing the Role

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Role Brief: Product Manager

What does a Product Manager do?

Sample Job Description

A Product Manager (PM) is the person who identifies the customer need and the larger business objectives that a product or feature will fulfill, articulates what success looks like for a product, and rallies a team to turn that vision into a reality. This includes managing the product throughout the Product Lifecycle, gathering and prioritizing product and customer requirements, defining the product vision, and working closely with engineering to deliver winning products.



Summary

Product Managers (PMs) identify market problems worth solving and then leads the group that will solve those problems.

Work Samples: Product Manager

Product Requirements Document (PRD)



Microsoft Outlook

Microsoft Outlook should build a subscription manager feature to streamline the user journey to clean and organize their inbox

Contributors: Bruce Liu

Last Updated: February 12, 2024

Problem Alignment |

As reliance on email as a communication platform grows, a prevalent challenge presents itself to the user experience. The **increase in promotional emails and subscriptions causes inbox clutter and disorganization**, leading to information overload and negatively impacting productivity and focus.

Through user interviews, a clear pain point has emerged: the difficulty users face in efficiently cleaning and organizing their inboxes. Specifically, **users find it time-consuming and cumbersome to unsubscribe from unwanted subscriptions and promotional emails**, leading to inbox clutter over time and the development of poor organization habits. This results in an unpleasant experience on the platform, and diminishes Outlook's product perception for current and potential new users.

High-Level Approach

This problem should be approached by considering how to minimize the friction and

Purpose of the PRD

Clear Communication

- Outline the specific features and functionality that will be built, which helps in setting the boundaries for what's included in the project

Alignment Across Teams

- Keep engineering, design, and other teams aligned with the product vision, scope, and priorities

User Needs and Use Cases

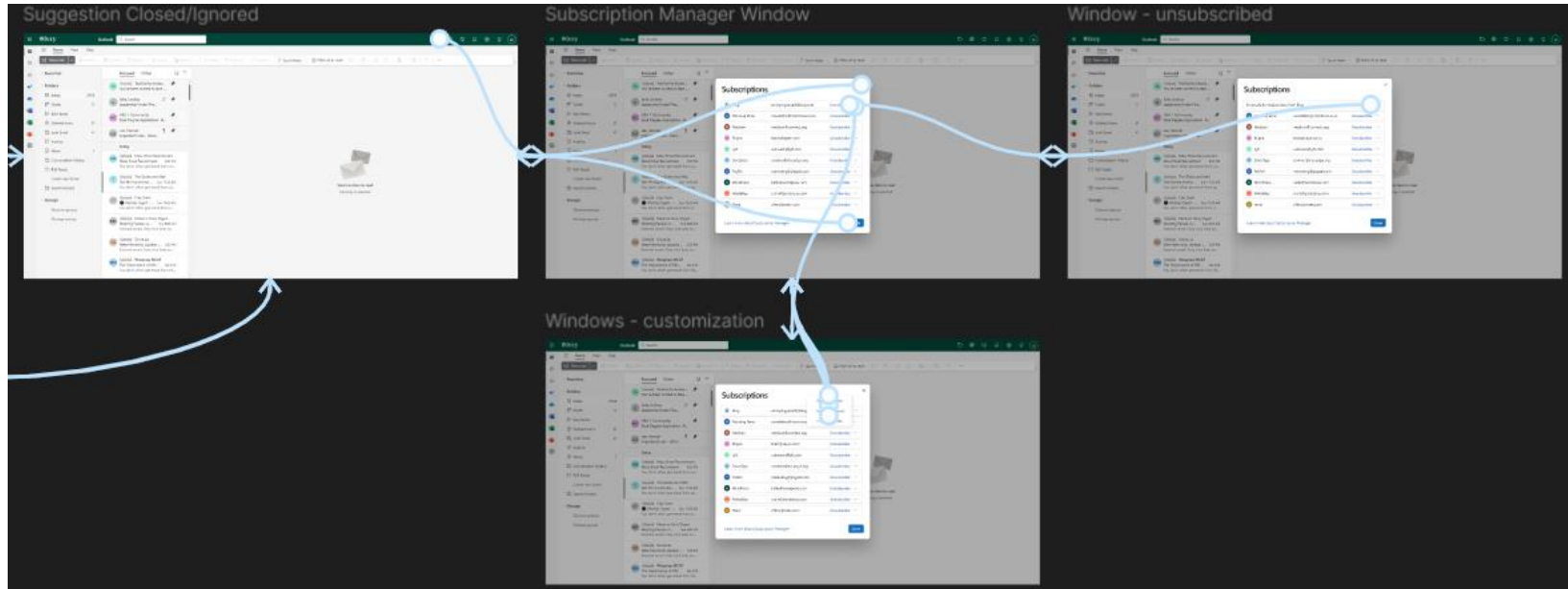
- Translate customer and market needs into actionable product features. This helps PMs prioritize features based on user problems

Success Metrics:

- Provide KPIs which guide the team in understanding how to measure the product's effectiveness once launched

Work Samples: Product Manager

High-Fidelity Mock-up / Prototype



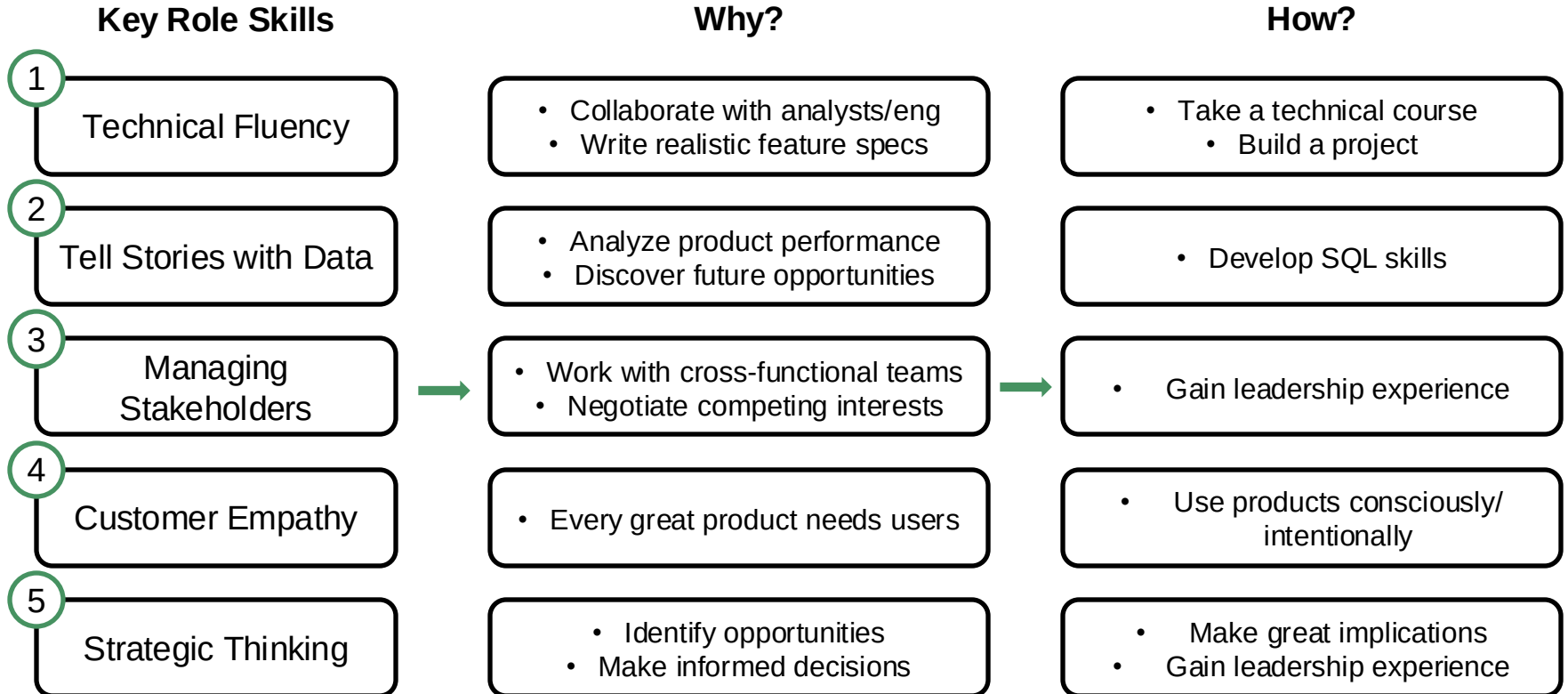


Section II: Key Skills and Competencies

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Top Skills: Product Management

Development and Presentation



Certifications and Bootcamps: Product Management

Cutting through the noise: are any worth it?



Experience	Description	Worth It? + Benefit
1 Ivey Product Fellowship	Free, 10-week bootcamp to learn PM skills, build a project, and pitch to industry leaders	YES!!!!!!!!!!
2 Riipen Project	PM project experience working with small businesses	
3 Entrepreneurship	'Own' a product and develop product skills	
4 Hackathons	Develop technical and soft skills – condensed product experience	



Section III: Early Career Progression

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Position Evolution

Responsibilities, timelines, and total compensation



US\$105,612

MEDIAN TOTAL COMP

US\$86.2K

25TH%

US\$133K

75TH%

US\$171.9K

90TH%

PRODUCT MANAGER | \$100,000 - \$250,000

2-5 years

First promotion

Product managers typically handle broader and more strategic initiatives, such as setting the product base and vision, overseeing the development and delivery of products, and leading a variety of teams to coordinate development efforts.

ASSOCIATE PRODUCT MANAGER | \$85,000 - \$180,000

2 years

Post graduation

APMs focus more on the day-to-day execution and delivery of the product roadmap, conducting critical pre-development tasks like market analysis, road mapping, and writing PRDs.

PRODUCT MANAGER INTERN | \$50,000 - \$100,000

4 months

HBA 1/HBA 2 summer

Interns work closely with a product manager to assist them with analyzing data, writing PRDs, defining features, facilitate sprints, collect user feedback, and more to build out a product/feature.

Position Evolution

Responsibilities, timelines, and total compensation



RBC Product Manager

Average Total Compensation

US\$81,889

Base Salary

US\$73,433

Stock Grant (/yr)

US\$0

Bonus

US\$8,456

Google APM

Average Total Compensation

US\$159,261

Base Salary

US\$140,682

Stock Grant (/yr)

US\$11,114

Bonus

US\$7,465

Google Product Manager

Average Total Compensation

US\$256,733

Base Salary

US\$172,880

Stock Grant (/yr)

US\$67,179

Bonus

US\$16,674

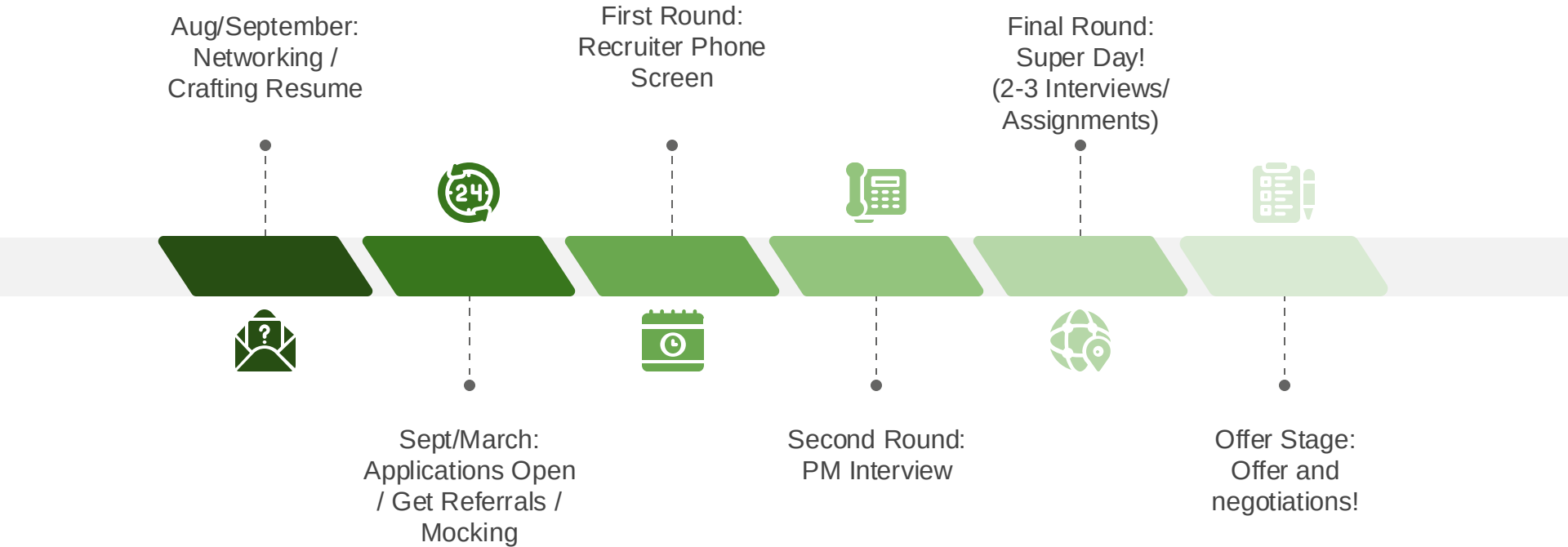


Section IV: Recruiting Timelines

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Recruiting Timelines

When should you apply?





Section V: Résumé, Interview, and Assessment Prep

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Resumé Breakdown

What goes into a good resumé?



Resume Breakdown

Name & Contact Information

Education

Professional Experience

Leadership Experience & Projects

Technical Skills & Interests

- What experiences can improve your chances of getting call backs from big tech:
 - Past PM experience
 - Past projects / entrepreneurship
 - Adjacent experience (SWE, tech consulting)
 - Highlight transferable skills
 - PM-related extracurriculars
- Examples of key skills:
 - User empathy, identifying pain points, user personas and journey maps
 - A/B testing and experimentation
 - Experience working with design, engineering, marketing, and operations teams
 - Prioritization and roadmap planning
 - Competitive analysis and business strategy
 - Understanding of sectors like fintech, blockchain, AI (relevant to role)

1

Recruiter Phone Screen

- **Interviewer:** Recruiter with survey-level knowledge of the role
- Phone screens are about structured communication. Think about it as an opportunity to demonstrate your strongest achievements.
- Other than normal behavioral questions, you may be asked:
 - Simple Product Sense: What's your favourite product and how would you improve it?
 - Knowledge check: How well do you know the company?

2 PM Interviews Round

- **Interviewer:** Junior or manager-level member of company
- Behavioral or Behavioral + PM
- Questions include:
 - Tell me about yourself + a few other behavioral questions
 - Product sense:
 - How would you improve Instagram stories?
 - Design an app for farmers in India
 - Make sure to research your interviewer to prepare good questions at the end

3

PM Interviews Round

- **Interviewer:** Senior Manager or VP of Product
- Consists of 2-3 back-to-back interviews
- Interview types include:
 - Product Sense
 - Estimation – How many restaurants are in San Fransisco?
 - Metrics – How would you measure the success of Whatsapp?
 - Investigation – Engagement just went down 5% WoW, as a PM on the team, what do you do?
 - Others
 - Strategy
 - System Design
 - Leadership/Coordination



Section VI: How to Stand Out in the Next Month and Year

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Your Roadmap For Success

Your Goal: Landing your first product role.



The Prep Work

- 1 Adapt your resume for PM the best you can.
- 2 Explore different PM roles by talking to upper years and industry professionals.
- 3 Figure out where your strengths/weaknesses in the PM role.



Build off your strengths and mitigate your weaker areas.

Prepare behavioural questions and reframe them for product management.

Land a PM role at a startup or small/mid-sized company. Typically involves cold outreach/networking.

Build Your PM Skillset

The areas you can differentiate yourself in



Product Sense

- Do a case study on a new product/improve a product and document it
- Read articles on successful products

Design

- Do a UX/UI case study and document it
- Experiment with design technologies

Data

- Learn SQL and/or Python
- Participate in datathons

Technical

- Code a project
- Lead a team of friends to launch a product

Leadership

- Apply for leadership roles at ANY club

Industry

- Read articles/thought leadership on latest trends in an industry
- Start writing your own articles

The best experience you can get is to **launch a product yourself.**

Meet your Workshop Leaders

It was a blast, let's stay in touch!



Sophia Ma
Prev. PM Intern at Mistplay

CS + Ivey HBA 2
I memorized 141 countries in 1h
[LinkedIn](#)
E: sma.hba2025@ivey.ca



Rachel Zhu
Prev. PM Intern at Zynga

CS + Ivey HBA 2
Taught in Kenya for 3 weeks
[LinkedIn](#)
E: rzhu.hba2026@ivey.ca



Bruce Liu
Prev. TPM Intern at Symend

CS + Ivey HBA 2
Survived a major tech layoff
[LinkedIn](#)
E: bliu.hba2025@ivey.ca



Grace Zhou
Prev. APM Intern at Coinbase
Prev. PM intern at Ripple

CS + Ivey HBA 2
Launched a cryptocurrency
[LinkedIn](#)
E: gzhou.hba2025@ivey.ca



Thank you!

Questions, next steps?



Technology Consultant: Beginner Skills for Recruiting and Career Growth

Delivered by Jarry Wu and Joyce Liu, Breakout Classroom 1140

Meet your Workshop Leaders

Glad to be here!



Jarry Wu

Tech Consultant at Konrad

Ivey HBA 2

*I slept on Joyce's couch for
all of first year*

LinkedIn: Jarry Wu

E: jwu.hba2025@ivey.ca



Joyce Liu

**Prev. Tech Consultant at EY,
Strategy at Bain**

Ivey HBA 2

*Jarry slept on my couch for all
of first year!*

LinkedIn: Joyce Liu

E: jliu.hba2025@ivey.ca

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2. [Day in the life](#)
3. [Early Career Progression](#)
4. [Recruiting Timelines](#)
5. [Resumé, Interview, and Assessment Prep](#)
6. [How to Stand Out in the Next Month and Year](#)



Section I: Introducing the Role

[Click here to return to table of contents](#)

Role Brief: Technology Consultant

What does a Tech Consultant do?



Job Description

Technology consultants are jacks-of-all-trades. Depending on personal skills and project needs a tech consultant could play the role of business analyst, project manager, software tester, or even developer. Consultants are often called upon when companies need guidance for large technology changes internally, such as new software or physical technology upgrades.



Summary

Technology consultants help fill gaps in client expertise, add more manpower to hefty technology projects, and guide the direction of once-a-decade technology upgrades within firms.

Day in the Life: Technology Consultant

What does a Technology Consultant do?



9AM: Start work, make coffee, grab a snack, check emails

9:30AM: Daily standup with supervisors + team

10AM-12PM: Client meetings

12-1PM: Lunch

1-2PM: Collaborate with colleagues on daily tasks

2-2:30PM: Meeting with mentor

2:30-4PM: Build client deliverables and slide decks

4-5PM: Coffee chat execs and other cool people :)

5:00PM: Leave office (hopefully)



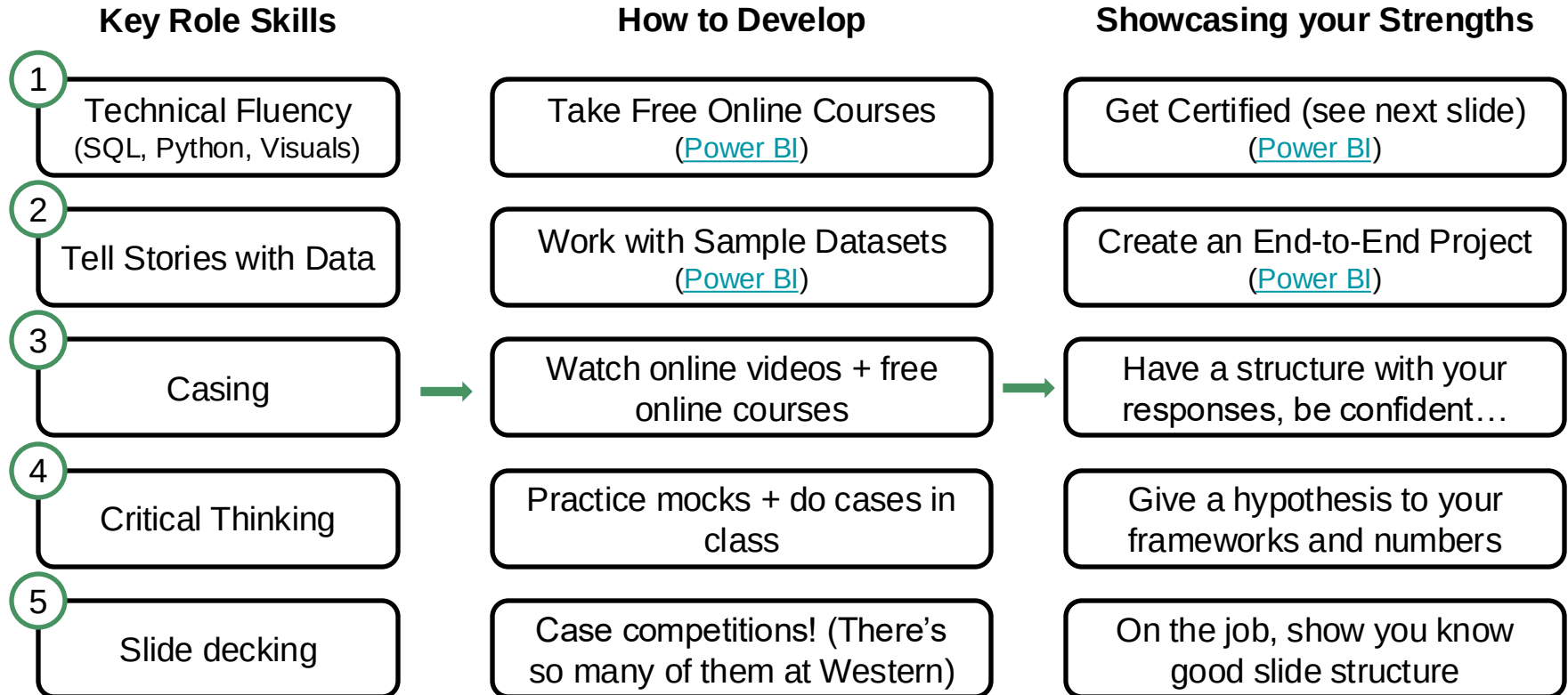


Section II: Key Skills and Competencies

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Top Skills

Data analysis, presentation, and problem-solving



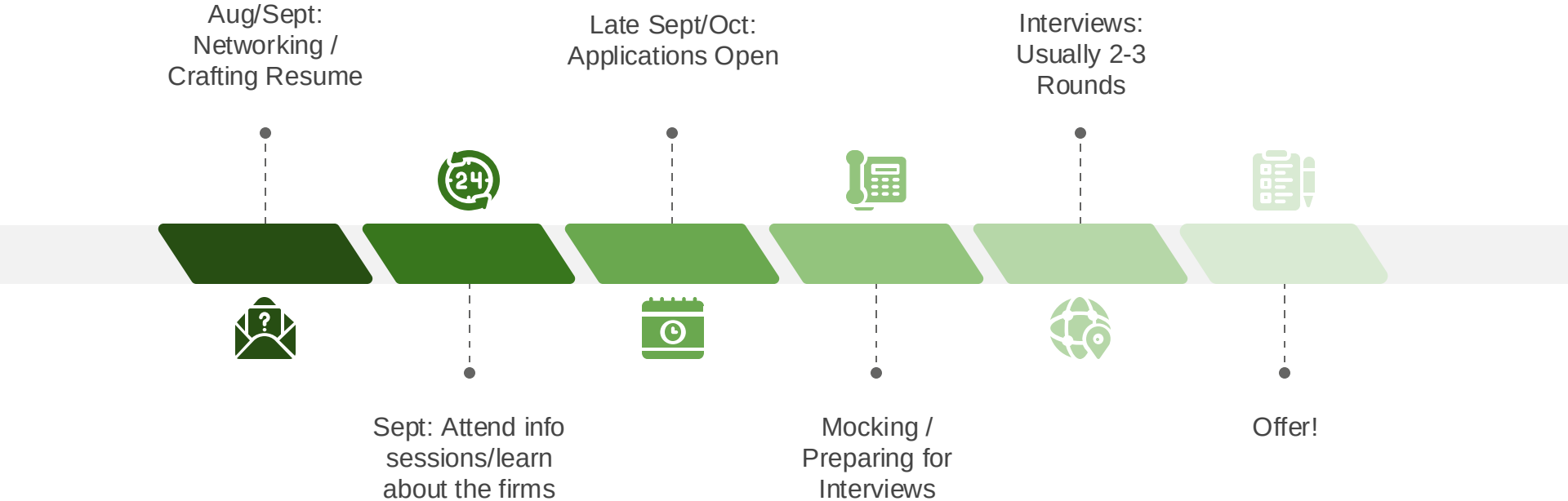


Section III: Early Career Progression

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Recruitment Timeline

Standard timeline for big tech consulting firms



Resume

How to get the recruiter's attention

Resume Breakdown

Name & Contact Information

Education

Professional Experience

Projects + Leadership experience

Technical Skills & Interests

- 3-4 accomplishment-oriented bullet points per work experience
- Most recruiters are not familiar with technical terms, so make it easy to understand what you did
- Highlight technologies you worked with, the impact you had (e.g. reduced rendering speeds by x%) and numerics (don't always have to be impact based, i.e. users, client net worth, size of dataset, etc.)
- Highlight technical skills mentioned in the job description
- Extracurriculars, clubs, and side projects that demonstrate passion for leadership
 - Highlight consulting-specific, transferable skills and experiences like working with clients, stakeholder management
 - Best experiences are the ones where you can talk about your specific contributions and impacts!

Interview Process

Typical interview timeline

1

Resume and Cover Letter Drop

- Make sure to customize the first and last paragraphs of your cover letter to highlight your unique interest in the firm
- If possible, name drop members of the firm that you have a genuine relationship with

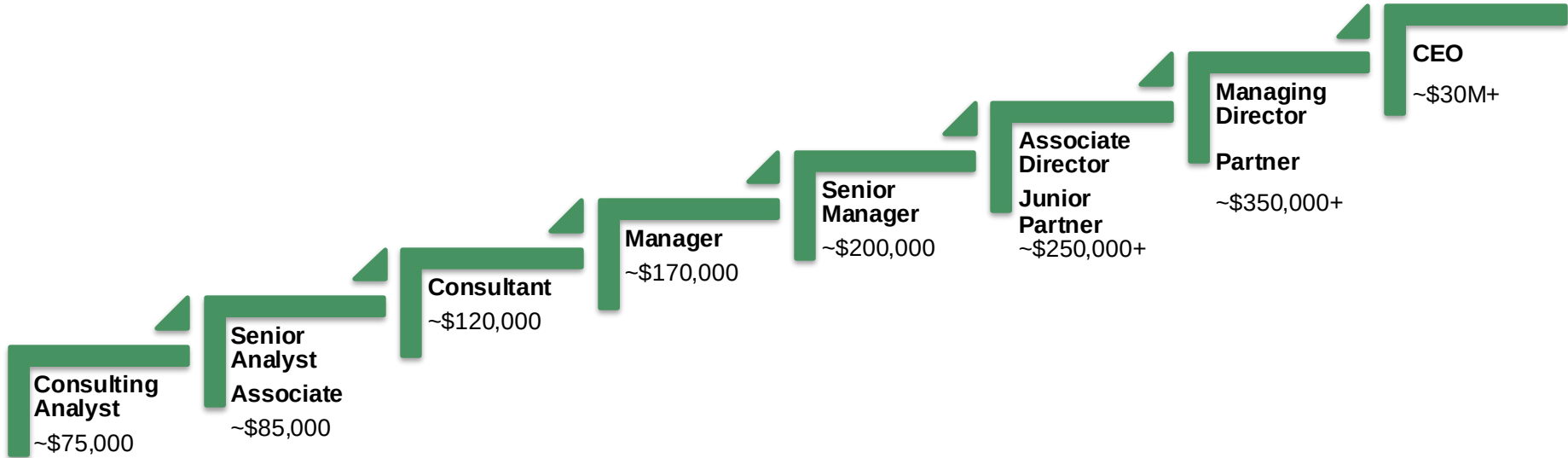
2

Interviews*

- Usually 2-3 rounds
 - **Round 1:** 2 interviews with Associates/Engagement Managers
 - **Round 2:** 2 interviews with Partners/Associate Partners
- A behavioural and case component (the weighting between the two differs based on the firm)
 - Interviewer vs Interviewee-led Cases
- Skills Tested
 - Charts and data
 - Mental math
 - Structure
 - Brainstorming/Solutioning
- Tips to stand out
 - Don't memorize a framework!
 - Be authentic and add human emotion
 - Try to understand specific nuances/lingo of the most common industries (e.g. CPG, pharmaceuticals, mining, etc.)
- *Note: Sometimes there's an online assessment/timed challenge prior to the interviews

Career Progression in Tech Consulting

A timeline for larger firms



Position Evolution

Responsibilities and timeline



SENIOR CONSULTANT – \$100,000

2-5 years

First promotion

Independently manage complex analysis projects, providing strategic recommendations to technical leads and senior management. Play a pivotal role in leading operational and strategic initiatives in the business unit.

STAFF CONSULTANT I/II – \$75,000

2-3 years

Post graduation

Support senior analysts and workstream leads in conducting process analyses, developing key artefacts (e.g. project management plan, feature backlog), and implementing process improvement initiatives.

CONSULTING INTERN – \$60-70,000

4 months

HBA 1/HBA 2 summer

Work alongside a BA team or a team of interns to define user requirements, analyze implications for implementation, and document decisions and changes as required.

Interview Process

Typical interview timeline

Prep Resources

- **Ivey Consulting Club**
 - Stay ahead of Ivey-specific application deadlines and gain access to case prep books, mentors, etc.
- **Rocket Blocks**
 - Overview of concepts, mental math drills, industry sprints, etc.
 - Checkout Ivey Career Management page on learn to figure out how to sign up for free
- **Hacking the Case Interview Youtube Channel**
 - Quick videos if you're prepping on a time crunch



Technology Consulting: Advanced Skills for Recruiting and Career Growth

Delivered by Rajah Chari, Izzie Pewarchuk, Ana Balteanu; Breakout Room 1220

Meet your Workshop Leaders

Glad to be here!



Rajah Chari

**Security Consulting Analyst
@ Accenture**

Comp Sci & Ivey HBA2

Lived in Mexico for 10+ years

LinkedIn

E: rchari.hba2025@ivey.ca



Ana Balteanu

**Technology Consultant @
Konrad Group**

Comp Sci & Ivey HBA2

I love olives

LinkedIn

E: abaltean.hba2025@ivey.ca



Izzie Pewarchuk

**SAP Technology Consultant @
EY**

Comp Sci & Ivey HBA2

I collect finger puppet magnets!

LinkedIn: Isabella Pewarchuk

E: ipewarchuk.hba2026@ivey.ca

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Section I: Introducing the Role

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Day in the Life: Technology Consultant

What does a Technology Consultant do?



9AM: Start work, make coffee, grab a snack, check emails

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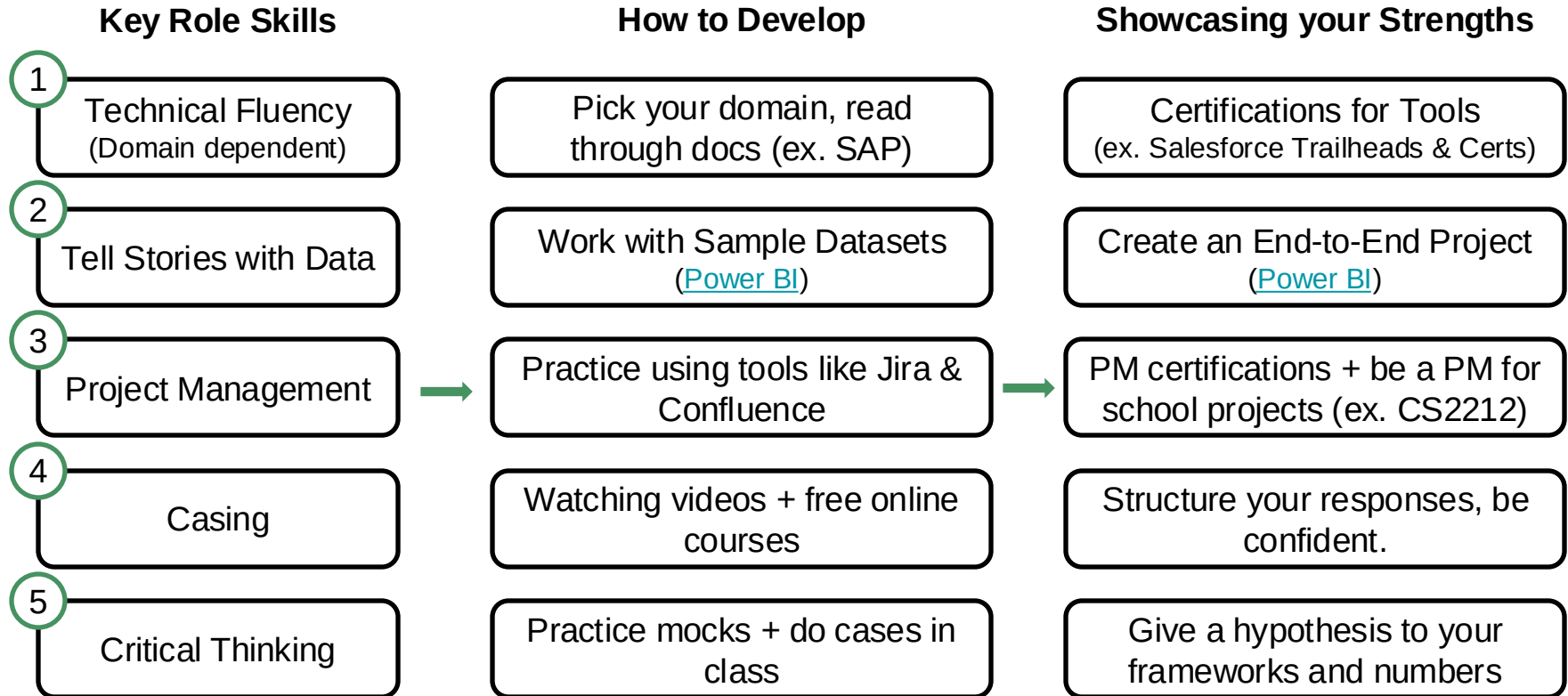


Summary

A lot of client interaction and teamwork to help the client achieve the best possible outcome (while learning a lot along the way).

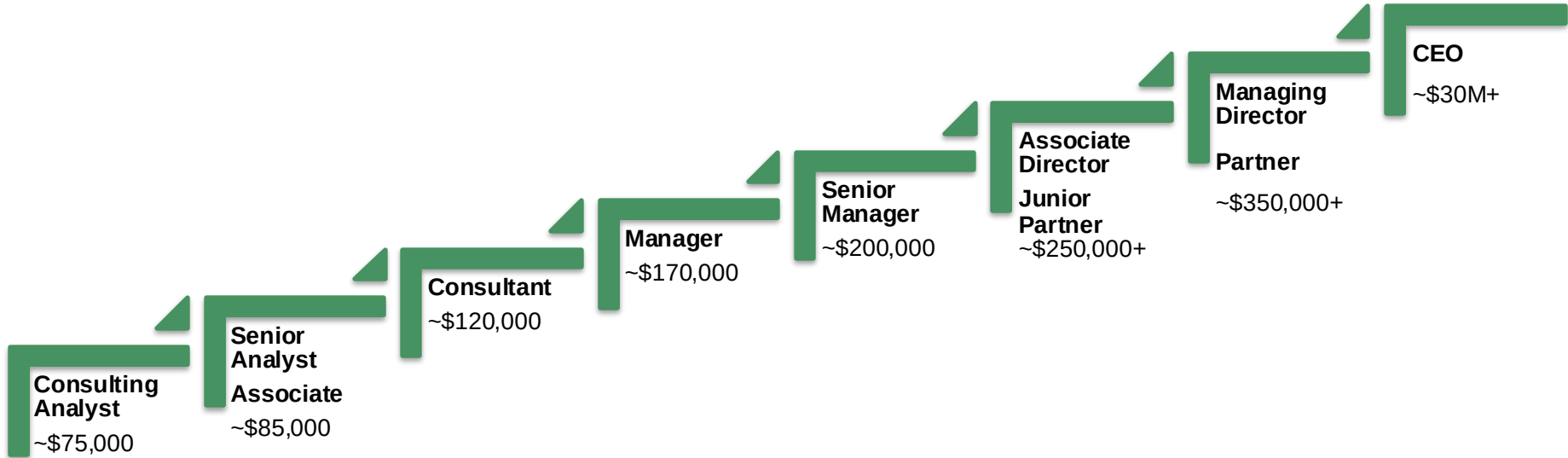
Top Skills

Data analysis, presentation, and problem-solving



Career Progression in Tech Consulting

A timeline for larger firms





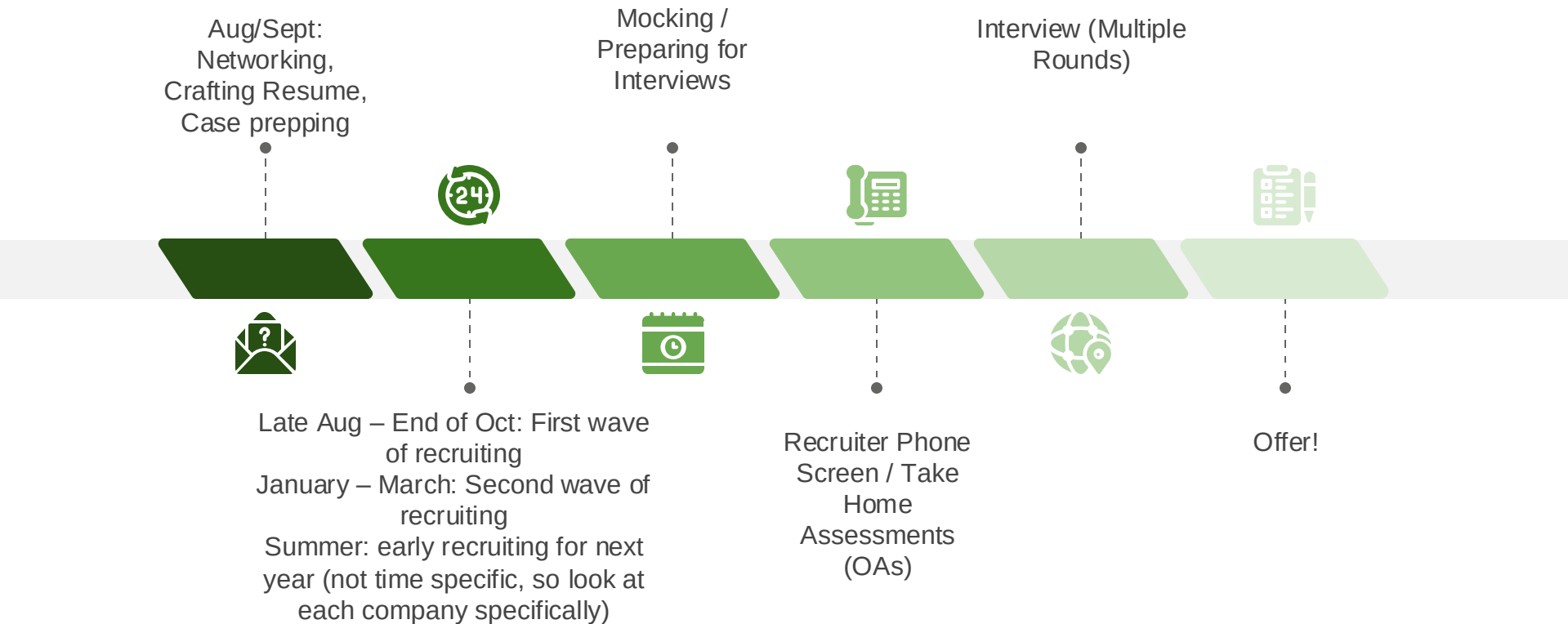
Section IV: Recruiting Timelines

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Recruiting Timelines

When should you apply?

Where to find postings?





Section III: Resume, Interview, and Assessment Prep

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Resume

How to get the recruiter's attention

Resume Breakdown

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 - Highlight consulting-specific, transferable skills and experiences like working with clients, stakeholder management
 - Best experiences are the ones where you can talk about your specific contributions and impacts!

Networking

How to get ghosted 🙄



Himanshu Patel · 1st
Manager, SecOps | CISM, OSCP, ECIH, CEH

SEP 1, 2023



Rajah Chari (He/Him) · 11:32 AM

Hi Himanshu! My name is Rajah with a great interest in cybers. Sam Armstrong recommended as I am looking to apply for a would appreciate some advice available for a quick call next



Rita Zurbrigg (She/Her) · 1st
Trust Enablement Lead at Google Cloud



Rajah Chari (He/Him) · 9:09 PM

Hi Rita! I am a computer science student at Western University with a passion for cybersecurity. I am



Alex Hart · 1st
IT Security Consultant

NOV 13, 2023



Rajah Chari (He/Him) · 9:02 PM

Hi Alex! I am a computer science student at Western University with a passion for cybersecurity. I am looking to apply to a few jobs in the field (including some at IBM) and would really appreciate some advice. Are you available for a quick call this/next week to chat?



Joseph Motha · 1st
Software Engineer @ Express



Rajah Chari (He/Him)

Hi Joseph! I am a computer science student at Western University with a passion for cybersecurity. I am



Rajah Chari (He/Him)

Hi Gareth, I am a computer science student at Western University with a passion for cybersecurity. I am looking to apply to a few jobs in the field (including some at Microsoft) and would really appreciate some advice. Are you available for a quick call this/next week to chat?

Gareth Pocock · 1st
Director of Compliance at Express

Samuel Zhang (He/Him) · 1st
PM @ Microsoft | cs & business '23'

OCT 14, 2023



Rajah Chari (He/Him) · 11:55 PM

Hi Samuel! I am a computer science student at Western University with a passion for cybersecurity. I am looking to apply to a few jobs in the field (including some at Microsoft in Seattle) and would really appreciate some advice. Are you available for a quick call this/next week to chat?



Valentina Giuliani (She/Her) · 1st
Campus Talent Acquisition Programs Manager

JUL 25, 2023



Rajah Chari (He/Him) · 11:21 AM

Hi Valentina! My name is Rajah Chari and I am a student at Western University that is passionate about cybersecurity. I wanted to learn more about the different roles PwC offers, your experience at PwC, and the recruitment process. Are you available for a quick Zoom call sometime next week to chat?



Sam Armstrong · 1st
Cyber Security Consultant

AUG 18, 2023



Rajah Chari (He/Him) · 5:31 PM

Hi Sam! I am a student at Western University (and a fellow WFN alum!) that has a passion for cybersecurity. I am looking to apply to a few positions in the field (including some at PwC) and would greatly appreciate some advice. Are you available for a quick Zoom call next week to chat?



Networking

Only takes one to work!



Rajah Chari (He/Him) • 10:09 PM

Hi Naeem! I am a computer science student at Western/Ivey with a passion for cybersecurity. I am looking to apply to a few jobs in the field (including some at Accenture) and would really appreciate some advice. Are you available for a quick call this/next week to chat?



Naeem Budhwani • 10:15 PM

Hey Rajah! Would love to help where I can. How about Friday AM or weekend?



Rajah Chari (He/Him) • 11:57 AM



Hi Andrew! I am a student at Western/Ivey with a passion for cybersecurity. I am looking to apply to a few positions in the field and would greatly appreciate some advice. Are you available for a quick Zoom call t week to chat?



Irew Hart • 11:58 AM

Rajah! Nice to meet you, definitely happy to chat! you give me a few dates and times that work for and we can set something up!



SURPRISE TECH CONSULTING CASE

ALWAYS BE ON GUARD



Airports of Things: Cloud Computing and IoT Devices in Travel Industry

Technicals

Technical cases and questions



Crafting Cases (Free!)

- Helpful resource for learning the frameworks of strategy consulting cases, basic business knowledge, and how to solve quantitative questions!

Rocketblocks + CaseCoach

- Good for drilling and practicing what you learn at CraftingCases!
- Also has videos that walk through the frameworks + quantitative questions
- Accessible through Career Management resources

ChatGPT

- Tech consulting interviews tend to be unique company to company; ChatGPT can be useful in coming up with specific questions that fit that specific interview format (just make sure to feed it enough information to do so)

Network

- Network to understand how interviews work at the company you are interviewing at! Tech cases differ so much depending on the company, so this is the most important prep you can do.

Cracking the PM Interview

- Great book with some advice for behavioural and technical questions – career management has copies!

Behaviourals

What can make you stand out

“Tell me about yourself”

- Talk about your education, past work experience, and how it's led to you wanting to work at this company
- Everything should tie together, don't bring up being a camp counsellor if it has nothing to do with working at Accenture
- Ex. “I started coding in high school because I found databases really interesting, which led me to pursuing a dual degree in computer science and business. I still find databases really interesting, and that's why I worked at X and now want to work here. Your company culture really aligns...”

CAR Framework

- Context → give an example of a situation you were in with that outcome: “In my finance final project, we were in groups and there was conflict...”
- Action → what was the action you took? How did it affect the situation?: “I suggested solving it by...”
- Resolution → what was the result of that action? What was the impact?: “We ended up coming to an agreement and doing well on the project. I learned...”
- Can use it for negative or positive situations; make sure to talk about what you learned and would change for negative situations
- Always good to structure so that it's easy for them to follow!

Behaviourals

What can make you stand out

Be personable

- Surprisingly, a lot of people don't know how to interview
- Don't memorize answers to questions, just memorize your experience history so that you are flexible to any question → think about scenarios that different experiences would align to
- Be authentic and personable, make sure that they get to know YOU and not "YOU", personality always is better than being rigid

Glassdoor is your friend

- If the company is big enough, most of the time you will be able to find their common interview questions online
- Good way to prep and know in advance what types of questions they will ask (helps you calm down too!)

RESEARCH THE COMPANY!!!

- **DO NOT** walk into an interview having no clue what a company does, what types of projects they work on, what role you are interviewing for, or not know about their company culture
- Interviewers can SMELL it and will know right away that you don't know, and it will come off as you not caring
- Learn about their culture, how it aligns with your values, what types of projects they work on and if it aligns with what you want, and what their day-to-day looks like → best way is to coffee chat!



Section IV: How to Stand Out in the Next Month and Year

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Coffee chat, coffee chat, coffee chat

Meet internal staff members



Prepare

- Research the person's background (LinkedIn, company site)
- Prepare 3-5 thoughtful questions
- Know your goal: Is it advice, insights, or networking?
- Bring a notebook to jot down insights (ask if it's OK to take notes, not everyone likes it)

Ice-Breakers

- How did you get into your current role?
- What's your advice for someone interested in [industry/role]?
- What's a challenge your team/company is currently facing?
- What skills do you think are essential for [role]?

Follow up

- Send a thank-you email within 24 hours
- Mention specific insights or advice they shared
- Offer to keep in touch or help in any way
- Connect on LinkedIn (if appropriate)

Opportunities

How to break in



Job Resources

- Recruiting Events
- LinkedIn Job Alerts + 12Twenty Alerts
 - Set it up so you can keep up to date with any Tech Consulting internships



Software Engineering: Beginner Skills for Recruiting and Career Growth

Delivered by Vivek Jariwala and Kyle Chen, Breakout Classroom 1120

Meet your Workshop Leaders

Glad to be here!



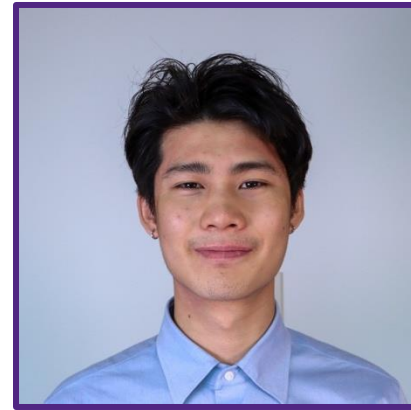
Vivek Jariwala

**Prev. Software Engineer
at Bond Brand Loyalty**

Software Eng & Ivey HBA2
*Sued by an Ivey MBA student
for a hackathon project*

[LinkedIn](#)

E: vjariwala.hba2026@ivey.ca



Kyle Chen

**Prev. Software Engineer
at RBC**

Comp Sci & Ivey HBA2
*Big fan of the
Toronto Maple Leafs*

[LinkedIn](#)

E: kylechen.hba2026@ivey.ca

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Section I: Introducing the Role

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Role Brief: Software Engineer

What does a Software Engineer do?



Sample Job Description

Software engineers use their knowledge of programming languages and systems architecture to design, develop, test, and maintain software applications. Whether it's creating a mobile app, developing complex algorithms, or working on large-scale systems, software engineers play a key role in turning ideas into working products. They collaborate closely with product managers, designers, and other engineers to understand user needs and develop efficient, scalable solutions. A strong grasp of problem-solving, logic, and collaboration is essential.



Summary

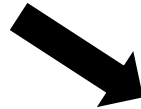
Software Engineers build, design, and maintain software.

Role Brief: Software Engineer

Different Streams of Software Engineering

When you think of Software Engineering...

There are many different streams that you can go into



Machine Learning Engineer

Full-Stack Developer

Mobile Engineer
(iOS, Android)

Data Engineer

Web Developer

Embedded Systems Engineer

Cloud Engineer

Work Samples: Software Engineering

Well obviously, coding . . .



Application.java - gs-spring-boot - Visual Studio Code

File Edit Selection View Go Debug Tasks Help

EXPLORER

- OPEN EDITORS
 - Welcome
 - Application.java complete\src\main\java\hello
- GS-SPRING-BOOT
 - complete
 - .mvn
 - .settings
 - bin
 - gradle
 - src
 - main
 - java
 - hello
 - Application.java
 - HelloController.java
 - test
 - .classpath
 - .project
 - build.gradle
 - gradlew
 - gradlew.bat
 - mvnw
 - mvnw.cmd
 - pom.xml
 - web.config
 - initial

Application.java

```
1 package hello;
2
3 import java.util.Arrays;
4
5 import org.springframework.boot.CommandLineRunner;
6 import org.springframework.boot.SpringApplication;
7 import org.springframework.boot.autoconfigure.SpringBootApplication;
8 import org.springframework.context.ApplicationContext;
9 import org.springframework.context.annotation.Bean;
10
11 @SpringBootApplication
12 public class Application {
13
14     public static void main(String[] args) {
15         SpringApplication.run(Application.class, args);
16     }
17
18     @Bean
19     public CommandLineRunner commandLineRunner(ApplicationContext ctx) {
20         return args -> {
21
22             System.out.println("Let's inspect the beans provided by Spring
23
24             String[] beanNames = ctx.getBeanDefinitionNames();
25             Arrays.sort(beanNames);
26             for (String beanName : beanNames) {
27                 System.out.println(beanName);
28             }
29
30         }
31     }
32 }
```

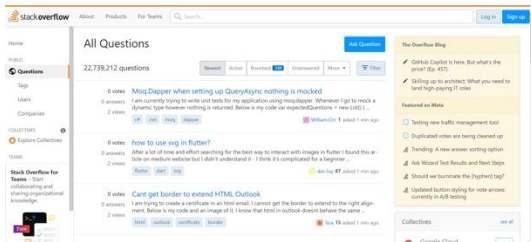
maven

Ln 15, Col 56 Spaces: 4 UTF-8 CRLF Java

Here is an example of how you can create a donut chart using Chart.js:

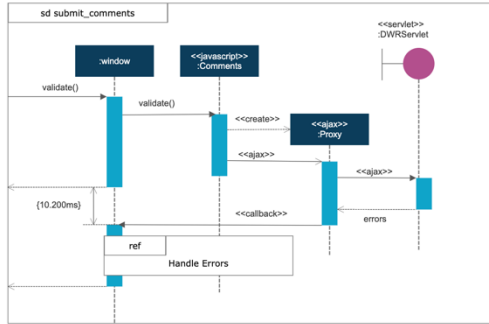
```
// Set the dimensions of the canvas element
var canvas = document.getElementById("myChart");
var ctx = canvas.getContext("2d");
var chartWidth = canvas.width;
var chartHeight = canvas.height;

// Create the data for the chart
var data = {
    labels: ["Red", "Blue", "Yellow", "Green", "Purple", "Orange"],
```



Work Samples: Software Engineering

But there's so much more (systems design, diagrams, test cases)



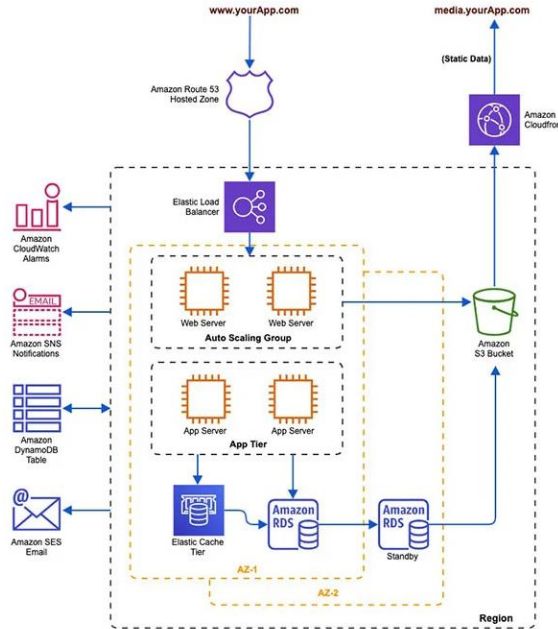
```
PS npm test

> learning-jest@1.0.0 test
> jest

PASS ./index.test.js
  FizzBuzz
    ✓ [3] should result in "fizz" (2 ms)
    ✓ [5] should result in "buzz" (1 ms)
    ✓ [15] should result in "fizzbuzz" (1 ms)
    ✓ [1,2,3] should result in "1, 2, fizz" (1 ms)

Test Suites: 1 passed, 1 total
Tests:       4 passed, 4 total
Snapshots:   0 total
Time:        0.586 s, estimated 1 s
Ran all test suites.
```

AWS 3-tier Architecture



Cloud Architecture – Designing and managing scalable applications and services on cloud platforms like AWS, Azure, or Google Cloud.

Test Cases – Writing scenarios to test how software functions under different conditions to ensure it works as expected.

Diagrams – Visual representations of system designs, workflows, or components to help communicate and plan software architecture.

Version Control – Systems like Git that track changes to code, allowing teams to collaborate and revert to previous versions if needed.

And so much more . . .

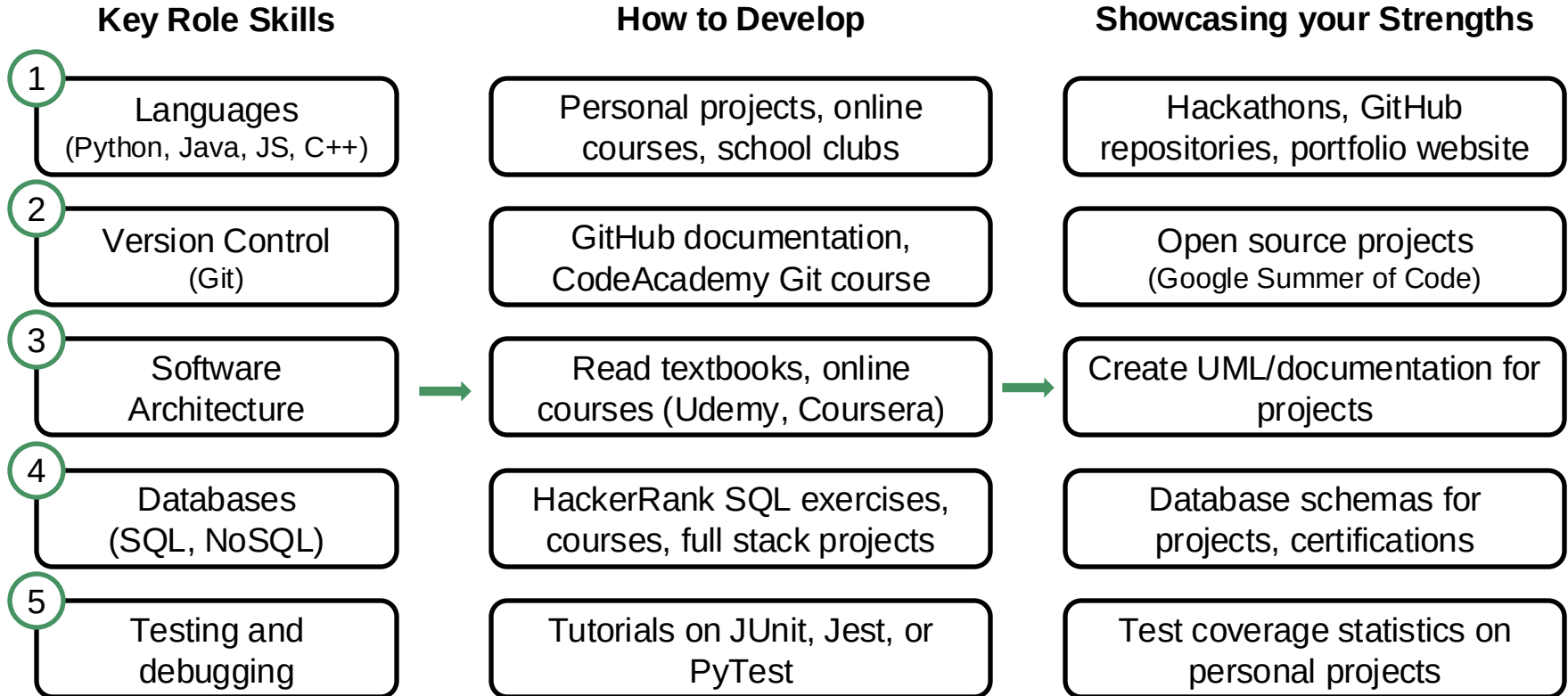


Section II: Key Skills and Competencies

[Click here to return to table of contents](#)

Top Skills: Software Engineer

Development and Presentation





Section III: Early Career Progression

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Position Evolution

Responsibilities and timeline



SOFTWARE ENGINEER II: \$125,000 – \$265,000

2-5 years

First promotion

Lead the development of complex systems, mentor junior engineers, and provide input on architecture and technical strategy across projects.

SOFTWARE ENGINEER I: \$70,000 – \$150,000

2-3 years

Post graduation

Design, develop, and maintain software applications, supporting senior engineers, conducting code reviews, and optimizing system performance.

SOFTWARE ENGINEERING INTERN: \$56,000 – \$77,000

4 months

Year 3/Year 4 Summer

Assist in real-world software development projects, help with debugging and testing, and gain hands-on experience with industry-standard tools and practices.

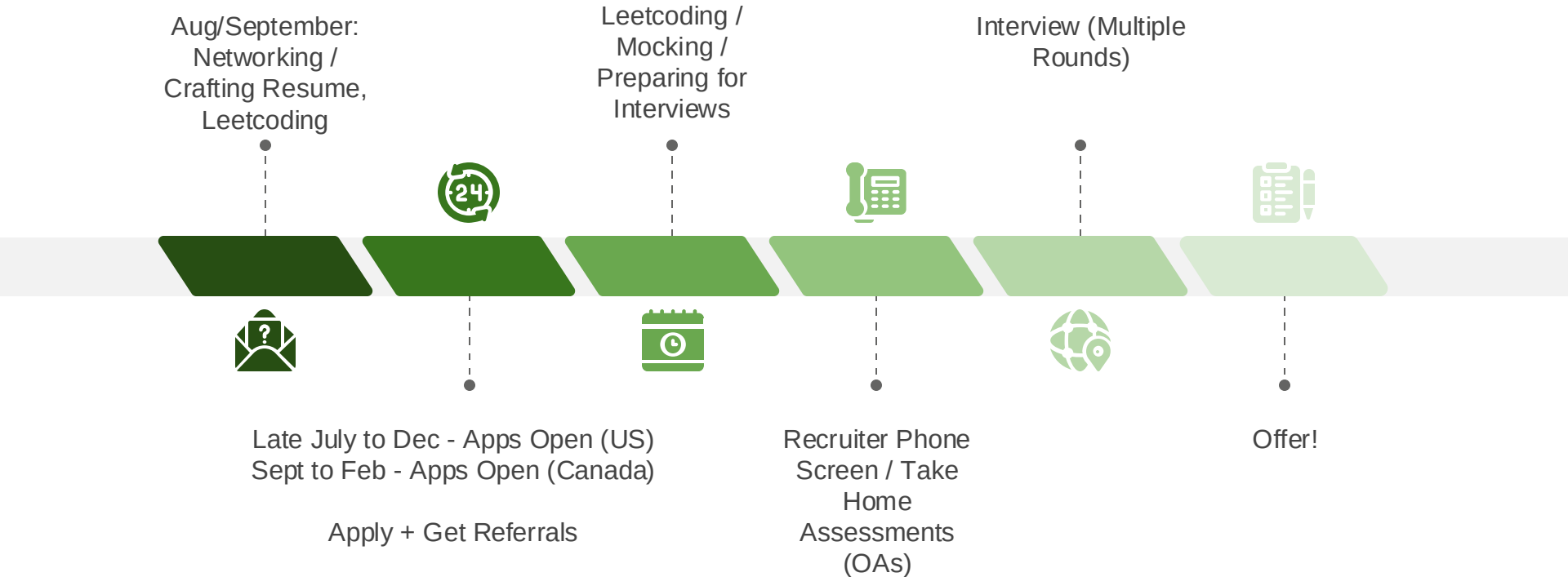


Section IV: Recruiting Timelines

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Recruiting Timelines

When should you apply?





Section V: Résumé, Interview, and Assessment Prep

[Click here to return to table of contents](#)

Resumé Breakdown

What goes into a good resumé?

Resume Breakdown

Name & Contact Information

Education

Technical Skills

Professional Experience

Leadership Experience & Projects

- 3-4 accomplishment-oriented bullet points per work experience
- Most recruiters are not familiar with technical terms, so make it easy to understand what you did
- Highlight technologies you worked with, the impact you had (e.g. reduced rendering speeds by x%) and numerics (don't always have to be impact based, i.e. users, client net worth, size of dataset, etc.)
- Highlight technical skills mentioned in the job description
- Extracurriculars, clubs, and side projects that demonstrate passion for tech and developing
 - Show that you love to code outside of the classroom - having your GitHub and website (if applicable) is great!
 - Despite the stigma, reframing school projects to sound like personal projects, especially when building out your portfolio is fine!

Resumé Breakdown

What template should I use?



Jake's Resume (LaTeX)

Jake Ryan

123-456-7890 | jake@osu.edu | [linkedin.com/in/jake](https://www.linkedin.com/in/jake) | github.com/jake

EDUCATION

Southwestern University <i>Bachelor of Arts in Computer Science, Minor in Business</i>	Georgetown, TX Aug. 2018 – May 2021
Blinn College <i>Associate's in Liberal Arts</i>	Bryan, TX Aug. 2014 – May 2018

EXPERIENCE

Undergraduate Research Assistant <i>Texas A&M University</i> - Developed a REST API using FastAPI and PostgreSQL to store data from learning management systems - Developed a full-stack web application using Flask, React, PostgreSQL and Docker to analyze GitHub data - Explored ways to visualize GitHub collaboration in a classroom setting	June 2020 – Present College Station, TX
Information Technology Support Specialist <i>Southwestern University</i> - Communicate with managers to set up campus computers used on campus - Assess and troubleshoot computer problems brought by students, faculty and staff - Maintain upkeep of computers, classroom equipment, and 200 printers across campus	Sep. 2018 – Present Georgetown, TX
Artificial Intelligence Research Assistant <i>Southwestern University</i> - Explored methods to generate video game dungeons based off of <i>The Legend of Zelda</i> - Developed a game in Java to test the generated dungeons - Contributed 50K+ lines of code to an established codebase via Git - Conducted a human subject study to determine which video game dungeon generation technique is enjoyable - Wrote an 8-page paper and gave multiple presentations on-campus - Presented virtually to the World Conference on Computational Intelligence	May 2019 – July 2019 Georgetown, TX

PROJECTS

Gitlytics <i>Python, Flask, React, PostgreSQL, Docker</i> - Developed a full-stack web application using with Flask serving a REST API with React as the frontend - Implemented GitHub OAuth to get data from user's repositories - Visualized GitHub data to show collaboration - Used Celery and Redis for asynchronous tasks	June 2020 – Present
Simple Paintball <i>Spigot API, Java, Maven, TravisCI, Git</i> - Developed a Minecraft server plugin to entertain kids during free time for a previous job - Published plugin to websites gaining 2K+ downloads and an average 4.5/5-star review - Implemented continuous delivery using TravisCI to build the plugin upon new release - Collaborated with Minecraft server administrators to suggest features and get feedback about the plugin	May 2018 – May 2020

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R
Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI
Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse
Libraries: pandas, NumPy, Matplotlib

<https://www.overleaf.com/latex/templates/jakes-resume/syzfjbzwjncs>

Resumé Breakdown

How do I write good resumé points?

- Used TypeScript to build a service that processes Excel files with customer data into user accounts.



- Lead production of an internal data import service, using **TypeScript** to parse Excel documents, converting them into 15,000 user profiles, reducing account creation times by 85%..

Interview Process

Steps 1 and 2: Online Assessment and Recruiter Phone Screen

1

Online Assessment (sometimes)

- **Platform:** HackerRank, CodeSignal, Codility, some random proprietary platform
- **While questions typically focus on LeetCode-style algorithms and data structures, the content might vary** depending on the role and the specific skills the company wants to assess. This may include tasks such as debugging, SQL queries, frontend implementation, or take-home projects.
- Can be proctored or unproctored (may still monitor user activity)

2

Recruiter Phone Screen (sometimes)

- **Interviewer:** Recruiter
- **Key question of the interview:** Does this person meet the job requirements?
- Phone screens are normally behavioural to assess whether you're a good fit for the team and culture
- You'll often be asked some basic technical questions as well

Interview Process

Steps 3 and 4: Technical and Behavioural Rounds

3

Technical (could be multiple rounds)

- **Interviewer:** Software engineer or another individual in a technical role
- Often LeetCode style questions but these can also be timed projects (i.e. 3-hour time limit to create a website or application), systems design questions (i.e. how would design Instagram's like feature?), technical questions (i.e. explain how to use Kubernetes to make your application scalable?)
- Search on Glassdoor/Reddit for what the interview looks like
- Structure: Questions → Base Case → Brute Force Solution → Optimize Solution → Code → Testing → Discuss tradeoffs and time/space complexity
- Things to review: Algorithms and Data Structures, Time/Space Complexity, System Design, whatever the interview could be about (coffee chats can be clutch for this)

4

Behavioural

- **Interviewer:** Manager or someone on your team
- **Key question of the interview:** Is this person a good fit?
- You could be asked some technical questions but most likely will be general questions about your experience and if you're a good fit (mainly culture, could be technical)



Section VI: How to Stand Out in the Next Month and Year

[Click here to return to table of contents](#)

How to Stand Out

Things to do within a month and a year



What to do in a Month

1) Create a portfolio of projects

2) Start a LeetCode Problem Set

3) Cold emailing (especially startups)

4) Join a tech club

5) Ask a professor to help with research

6) Get a certification

What to do in a Year

1) Build a project you're passionate about

2) Western Tech Alumni Summer Program

3) Participate in hackathons and network

4) Create a startup/business around tech

5) Leadership position in tech club

6) Social media



Section VII: LeetCode Example

[Click here to return to table of contents](#)

Interview Process

Walking through a LeetCode problem



242. Valid Anagram

Solved

Easy Topics Companies

Given two strings `s` and `t`, return `true` if `t` is an **anagram** of `s`, and `false` otherwise.

Example 1:

Input: `s = "anagram", t = "nagaram"`

Output: `true`

Example 2:

Input: `s = "rat", t = "car"`

Output: `false`

Constraints:

- `1 <= s.length, t.length <= 5 * 104`
- `s` and `t` consist of lowercase English letters.

Follow up: What if the inputs contain Unicode characters? How would you adapt your solution to such a case?

Discuss with someone sitting beside you how you would approach this problem! Feel free to use pseudocode or any language you know!

<https://leetcode.com/problems/valid-anagram/>

Interview Process

Resources for Interview Preparation



LeetCode: <https://leetcode.com/>

NeetCode: <https://neetcode.com/>

AlgoMap: <https://algomap.io/>

System Design – Gaurav Sen (YouTube)

And of course ChatGPT when you don't know the answer :P



Thank you!

Questions, next steps?

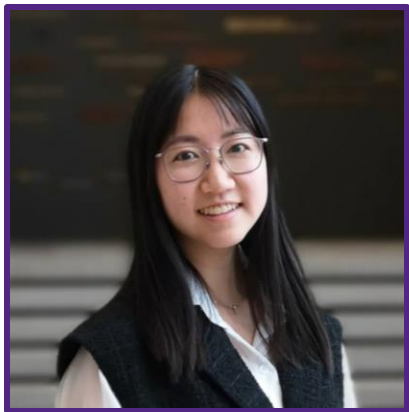


Software Engineering:

Delivered by Joy Zheng and Jeffrey Li, Breakout Classroom 1210

Meet your Workshop Leaders

Glad to be here!



Joy Zheng

Prev. Software Development

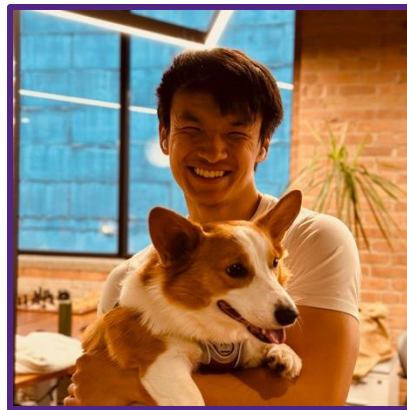
Engineer @ Amazon

Comp Sci & Ivey HBA2

Big Android supporter

[LinkedIn](#)

E: jzheng.hba2026@ivey.ca



Jeffrey Li

ML Engineer @ Cohere

Ivey HBA2

*Will go fishing after this
workshop. 🐟*

[LinkedIn](#)

E: jeffre yli.hba2025@ivey.ca

Table of Contents

Click on the links below to skip to each section.

1. [Introducing the Role](#)
2. [Recruiting Insights](#)
3. [Resume, Interview, and Assessment Prep](#)
4. [How to Stand Out in the Next Month and Year](#)



Section I: Introducing the Role

[Click here to return to table of contents](#)

Day in the Life: Software Engineer

What does a Software Engineer do?

9AM: Start work, push code review, check emails and Slack

9:30AM: Commute to office

10AM: Daily team standup

10:30-12PM: Head down work

12-1PM: Lunch

1-2PM: Head down work

2-2:30PM: Meeting with mentor

2:30-3PM: Cafe run

3-4PM: Test, fix errors, push code review

4-5PM: Chat with interns

5-5:30PM: Continue testing, fix errors

5:30PM: Leave office (hopefully)

~~9:00 PM: Run More Tests~~



Summary

Design, develop, and maintain software while writing documentation and improving operational efficiency.

3 Contrastive Policy Gradient

3.1 General objective

Contrastive approach requires pairs of generations, which do not need to be ranked as in RLHF. For a pair of independent generations (y, y') , we introduce the following sample loss:

$$\begin{aligned} \ell_{\text{CoPG}}(y, y'; \pi) = & \quad (5) \\ & \left(R_{\beta/2}^{\pi}(y) - R_{\beta/2}^{\pi}(y') \right) \ln \frac{\pi(y)}{\pi_{\text{ref}}(y)} \\ & + \left(R_{\beta/2}^{\pi}(y') - R_{\beta/2}^{\pi}(y) \right) \ln \frac{\pi(y')}{\pi_{\text{ref}}(y')}. \end{aligned}$$

This can be seen as a weighted log-likelihood, where the weight is the reward of the generation contrasted with the reward of an independent generation, and its symmetric.

Let μ_1 and μ_2 be some independent distributions (for example underlying a dataset of pairs of generations), that do not need to be known analytically (contrary to policy gradient with importance sampling), and that can

the other distribution. This loss is supervised-friendly, as it does not involve sampling from the trained policy.

The natural question is whether maximizing this objective L solves the intended problem (2), and thus maximizes any language scores. The answer is positive, and we deferred all proofs to Appx. A:

Theorem 1 (CoPG solves the right problem). *Assume that π_{ref} , μ_1 and μ_2 all have the same support. Then, the unique maximizer of $L(\pi)$, defined Eq. (6), is $\pi_*(y) \propto \pi_{\text{ref}}(y) \exp \frac{R(y)}{\beta}$, which is also the unique maximizer of $J(\pi)$.*

To shed more light on the relationship to policy gradient, let's consider the gradient of $L(\pi)$. By simple calculus (taking care of the fact that $R_{\beta/2}^{\pi}$ does depend on π), one obtains:

$$\begin{aligned} \nabla L(\pi) = & \mathbb{E}_{y \sim \mu_1} \left[\left(R_{\beta/2}^{\pi}(y) - \overline{R_{\beta/2}^{\pi}} \right) \nabla \ln \pi(y) \right] \\ & + \mathbb{E}_{y' \sim \mu_2} \left[\left(R_{\beta/2}^{\pi}(y') - \overline{R_{\beta/2}^{\pi}} \right) \nabla \ln \pi(y') \right]. \quad (7) \end{aligned}$$

When compared to Eq. (3), the classic policy gradient with baseline, we obtain a sum of two policy-like gradients, however with striking differences. First, the expectation is not according to the learnt policy π , but according to either μ_1 or μ_2 , meaning that it can be understood as a sound off-policy policy gradient. Second, there is a

```

92 class RPOMinizord(Minizord, MinizordCheckpointner):
177     return logits_preprocessing.extract_logprobs_from_logits(output, mask_key)
178
179
180 # | _ _ _ _ _ ( _ _ _
181 # | _ | _ / _ _ | | _ ' _
182 # | _ | _ | ( _ | | | | |
183 # | _ | _ | _ _ | _ | | |
184 #
185
186 def train_step(
187     self, step: int, data_iterator: Iterator, timeout: Optional[float] = None
188 ) -> Tuple[cheX.ArrayTree, float, cheX.ArrayTree]:
189     """Performs a single training step of the model.
190
191     Args:
192         step (int): Current training step.
193         data_iterator: Iterator that shall provide the training batch.
194         timeout (Optional[float]): Optional timeout for ray.get in seconds, if exceeded, an exception is raised. Set to None
195
196     Returns:
197         batch (Dict), loss (float), grad_norms (Dict)
198     """
199     with self._global_timer("batch_retrieval"):
200         batch: Any = next(data_iterator)
201
202     # Separate executor call for reference lprobs - memory optimisation
203
204     with self._global_timer("get_logprobs_from_logits_timer"):
205         batch["ref_lprobs"] = self._lprobs_remote_fn(self._nested_model_states["reference"], batch, mask_key="prompt_mask")
206         batch["ref_prefix_lprobs"] = self._lprobs_remote_fn(
207             self._nested_model_states["reference"], batch, mask_key="prefix_mask", timeout=timeout
208         )
209
210     # Note: When using multiple states, the same data structure will be provided to the update_fn
211     loss: Any, aux_metrics: Any = self._train_remote_fn(
212         model_states=self._nested_model_states,

```

...



runs.summary["summary_metrics"]

estimator	Train_Level_1_F	Partial_Complet	GSM8K_Pass_Ra	Math_Pass_Rate	MATH_Algebra	MATH_Counting	MATH_Geometr	MATH_Intermed	MATH_Number_	MATH_Prealgebi	MATH_Precalcul
FaxServer: pref- reasoning/	0.715	0.5875	0.7968	0.2529	0.405	0.245	0.165	0.12	0.185	0.535	0.115

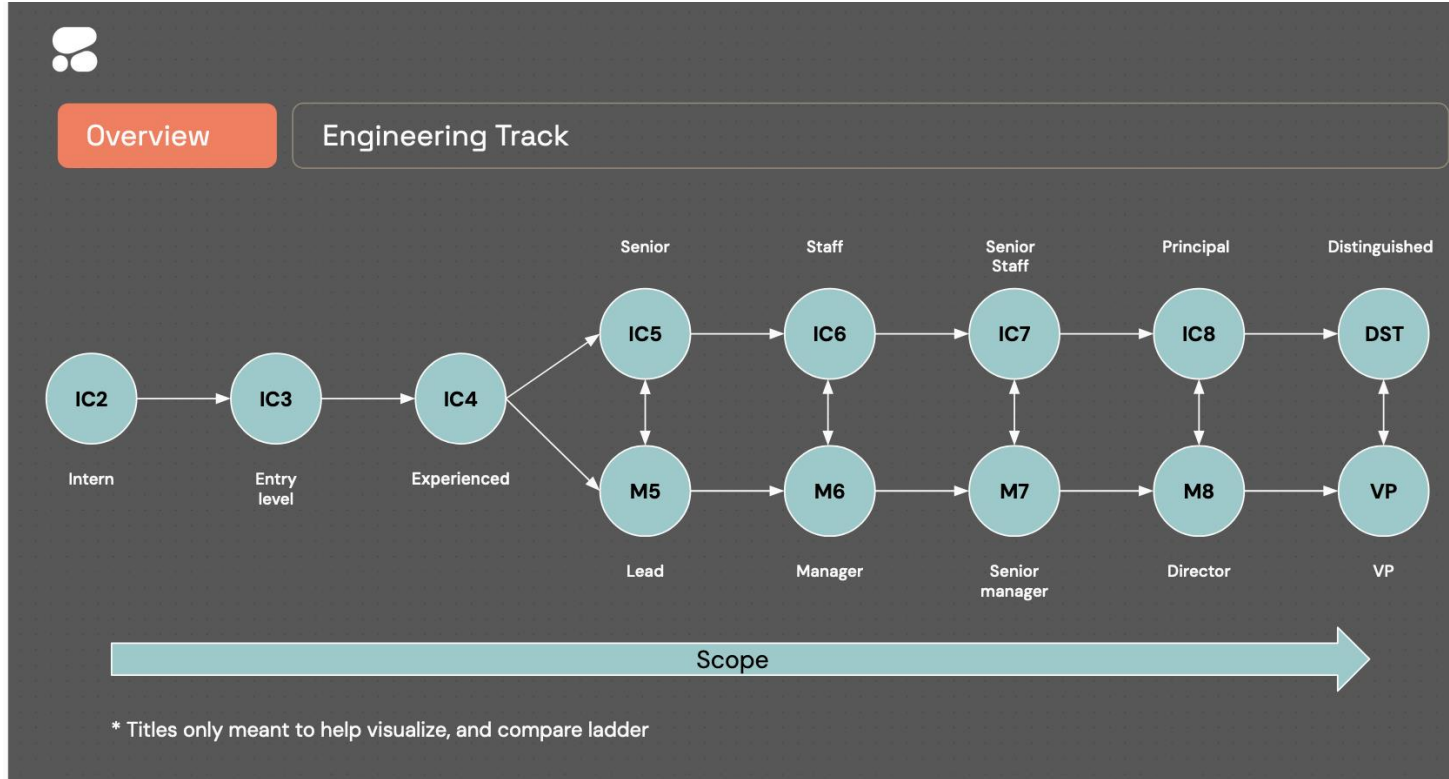
Top Skills: Software Engineer

How to be a good engineer



Position Evolution: Individual Contributor & Manager

Responsibilities and timeline



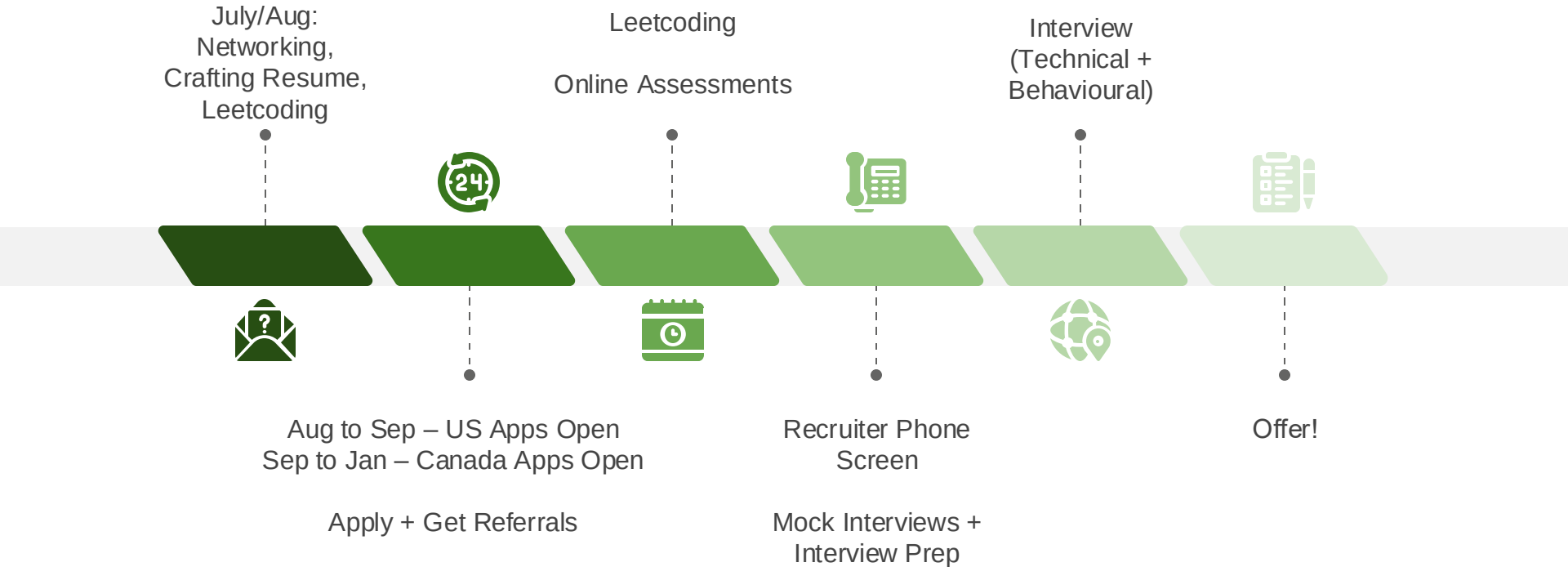


Section II: Recruiting Insights

[Click here to return to table of contents](#)

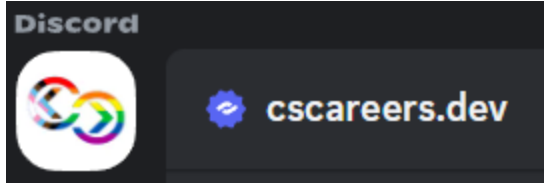
General Timeline

When should you apply?



Application Strategies

Where to find postings?



<https://discord.gg/cscareers>



Summer2025-Internships



- APPLY EARLY!!!
 - Ask past interns or recruiters for their application timelines
- Only include a cover letter if the posting explicitly looks for one
- Featured on 12Twenty:
 - RBC Amplify (due October 1)
 - Labatt (due October 31)
 - Dayforce (due January)
 - IBM (due February)

Networking

How to get ghosted



Christopher Tenn (He/Him) · 1st
Senior Specialist, Talent Acquisition at Konrad Group & BrainStation

APR 19, 2023



Jeffrey Li (He/Him) · 3:42 PM

Hi Chris! I'm a student at Western with software development experience working on startup, freelance and industry projects. I've recently been looking to try something new, Konrad's portfolio caught my attention. Would you be open for a quick chat about culture and



Gagan Gill

Write a message...



KB Bowles



KB Bowles (He/Him) · 1st
Talent @ Abridge | ML + Engineering

MAY 12, 2023



Jeffrey Li (He/Him) · 4:37 PM

Hi Katie! This past year, while studying ML at Western, I won two national AI Challenges hosted by IBM and Microsoft, and previously researched novel mathematics results at Ottawa U. I'm currently looking for a summer internship, I really believe the work you are building at Assembly is the future.

Write a message...



Dona Danhundo



Dona Danhundo · 1st
Recruiter

APR 3, 2023



Jeffrey Li (He/Him) · 9:16 PM

Hi Dona! This past year, while studying ML at Western, I won two national AI Challenges hosted by IBM and Microsoft, and previously researched novel mathematics results at Ottawa U. I'm currently looking for a summer internship, I really believe the work you are building at Microsoft is the future.



Ram Sriharsha
Available on mobile



Ram Sriharsha (He/Him) · 1st
CTO at Pinecone

MAR 29, 2023



Jeffrey Li (He/Him) · 5:30 PM

Hi Ram! I am leading development for a university team working on Long Term Memory applications for education - winning Microsoft's AI for Accessibility Challenge this past week. Pinecone has been an amazing fit for us, would love to connect for your expertise in this field

Write a message...



Kazakis

on mobile

Earning Content Development at IBM

APR 7, 2023

Li (He/Him) · 1:30 PM

tern, I
atics
ner
be a

filled.
year in



Section III: Resume, Interview, and Assessment Prep

[Click here to return to table of contents](#)

Resume

How to get the recruiter's attention



Resume Breakdown

Name & Contact Information

Education

Professional Experience

Projects & Leadership Experience

Technical Skills & Interests

- 3-4 outcome-oriented points per work experience
 - Explain WHY your work was important, HOW you did it, and quantify the impact
 - Metrics – processing speed, time savings, # of users, size of dataset, client net worth
- Ensure your writing is easy to understand for a non-technical audience
- Tailor your experiences and technical skills to the job description or role (depth > breadth if it's specified)
- Demonstrate your passion for tech and developing
 - Have an active GitHub and updated website
 - Personal projects and extracurriculars can show your interests beyond coding
 - Many avenues – Open source, CS research, fellowships, certifications, courses

SURPRISE LEETCODE

ALWAYS BE ON GUARD



Technicals

Online assessments & technical interview prep

- Be familiar with Data Structures + Algorithms and common patterns
 - Online course - Grokking the Coding Interview
- Read 3rd + 4th year DS&A books
 - A lot of Leetcode Hards are just variations of 3rd/4th year concepts
 - Popular book - Cracking the Coding Interview
- Practice consistently – Leetcode (Blind 75), CodeSignal, HackerRank, OAs
 - Watch Neetcode for solutions
- Search for company-specific OA and interview questions on Glassdoor, Reddit, Leetcode, [GitHub](#)
- Do mock interviews!



Behaviourals

What can make you stand out

BEFORE

- Do your research on the company and role -> Demonstrate cultural fit and alignment with leadership principles
- Have an elevator pitch -> Show why you do what you want to do, how you got to where you are, and how you would add value to the company

DURING

- Be authentic & personable -> They want to see how it's like working with you
- For technical questions, communicate your thoughts clearly and collaborate with your interviewer (treat it as a conversation)
- Ask thoughtful, non-searchable questions

AFTER

- Send a follow-up email (and manifest)
- Take each opportunity as a learning experience



Section IV: How to Stand Out in the Next Month and Year

[Click here to return to table of contents](#)

Building Internal Tools

Best Way To Gain Experience

My family generated pictures me from
interacting with or producing depictions
of violence. Is there something else I can
help you with?

could you write a paragraph about a
family picking flowers?

Sure! Here's a paragraph about a family
picking flowers:

cool thanks

You're welcome! Is there anything else I
can help you with?

Skip >

9:41

Welcome Back ...

All Time Statistics

👍 20 🗑️ 70 🍷 2 🎮 23

Daily Usage

Bar chart showing usage from 2012 to 2017. A green circle indicates a 3 Day Streak.

The Punnett square is a tool used to determine the probability of a particular genotype. It is a tabular summary of possible combinations of maternal alleles with paternal alleles. The Punnett square is a visual representation of...

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The Punnett square is a tool used to determine the probability of a particular genotype. It is a tabular summary of possible combinations of maternal alleles with paternal alleles. The Punnett square is a visual representation of...

Write code that solves this problem.

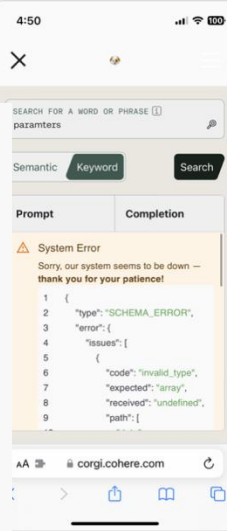
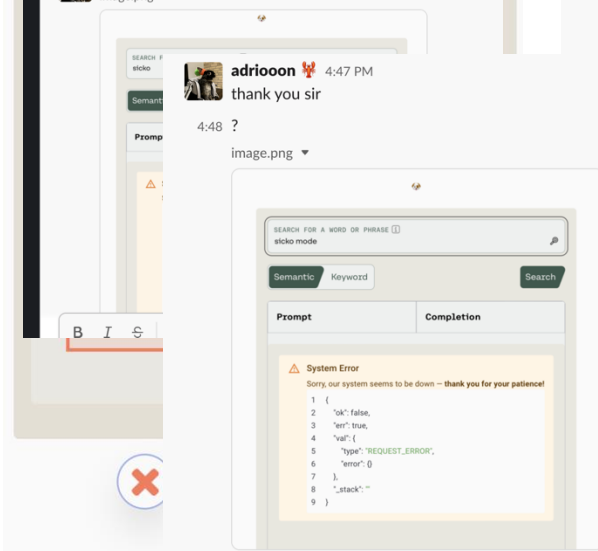
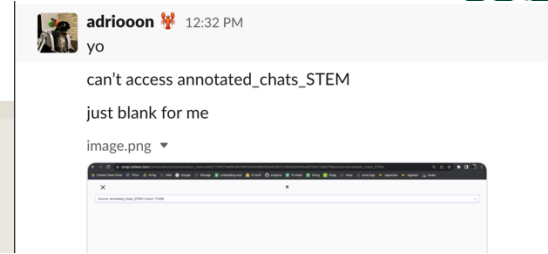
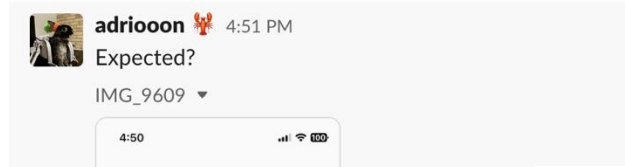
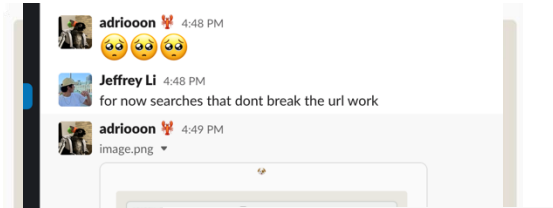
one 100 gallons of fuel are purchased
at \$ 0.70 per gallon . the fuel is
consumed at a rate of \$ 0.40 worth of
fuel per hour . at this rate , how many
hours are required to consume the 100
gallons of fuel ? $n0 = 100.0$ $n1 = 0.7$ $n2 = 0.4$ $n3 = 100.0$

A possible solution:

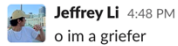
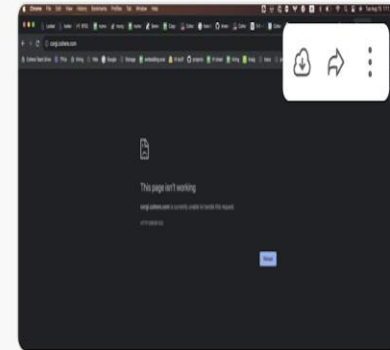
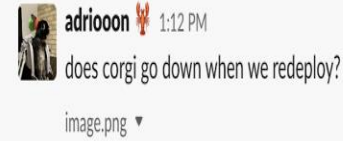
$n0 = 100.0$
 $n1 = 0.7$
 $n2 = 0.4$
 $n3 = 100.0$
 $t0 = n0 * n1$
 $answer = t0 / n2$

Building Internal Tools

Best Way To Gain Experience



it cuz we're uploading new data?



Opportunities

How to gain experience



- Project-based opportunities (Riipen, Venture for Canada)
- Federal Student Work Experience Program
- Canada Summer Jobs Bank
- MLH Fellowship
- Hackathons/Conferences (sometimes have on-site interviews, accelerated processes, email lists, or you can network for an interview)
- Y Combinator - Post for founding engineer roles (difficult to get but a good way to break into SF)
- Cold emailing founders/startups
- If you build a product, someone will do all customer calls, and pitching a working app is one of the best ways to show how to build → Get a business friend (Ray)
- Grinding technicals and getting better at OAs
- Join Western Tech Alumni (WTA) cohort in the summer
 - Receive referrals to 15+ big tech companies if you complete the homework and pass a mock interview



Thank you!

Questions, next steps?